The Cohomology Proposal for Anomalies and SPT

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Abstract

These notes organize the group-cohomology proposals for gauge anomalies and SPT phases around anomaly inflow. Sorting $(d+1)$-dimensional response functionals by whether they depend on the bundle class alone and whether their couplings are forced to be discrete separates $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ from Free$[H^{d+2}(BG;\mathbb{Z})]$, reproducing the global–perturbative division. The two sectors assemble into a single differential-cohomology group $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z})$. The limitations that survive the assembly are those of the classifying group, and they locate the exit from the ansatz in bordism. For finite G only the torsion sector survives; there, gauging an SPT phase yields a Dijkgraaf–Witten theory, and twisted gauging amounts to stacking an SPT before gauging.

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Conventions

Dimensions and manifolds. Throughout, d denotes the *spacetime* dimension of the theory whose anomaly is under study. Three dimensions occur, and the same cohomology groups appear in more than one of them:

symbol	dimension	role
$\mathcal{M}$	d	spacetime of the theory carrying the anomaly; also the spacetime of an SPT phase in §2
$\mathcal{Y}$	$d+1$	the inflow bulk: an invertible response theory whose boundary is $\mathcal{M}$
$\mathcal{X}$	$d+2$	auxiliary manifold with $\partial\mathcal{X} = \mathcal{Y}$, used only to *compute* a Chern–Simons action

A superscript is written whenever the dimension matters, e.g. $\mathcal{M}^d$.

Degrees. For discrete G one has $BG = K(G,1)$ and hence $H^n(BG;M) = H^n_{\mathrm{grp}}(G;M)$ in every degree. Degree shifts always come from a change of coefficients from $\mathbb{R}/\mathbb{Z}$ to $\mathbb{Z}$: either Wigner's isomorphism $\mathcal{H}^n_B(G;\mathbb{R}/\mathbb{Z}) \simeq H^{n+1}(BG;\mathbb{Z})$ for the Borel cohomology of a compact Lie group, or the Bockstein $H^n(BG;\mathbb{R}/\mathbb{Z}) \simeq H^{n+1}(BG;\mathbb{Z})$ for finite G, both for $n \geq 1$. The rule of thumb for where an object lives:

$$\begin{aligned} \lambda \in H^n(BG;\mathbb{R}/\mathbb{Z}) &\ \rightsquigarrow\ \text{a topological action on an } n\text{-manifold,} \\ \lambda \in H^n(BG;\mathbb{Z}) &\ \rightsquigarrow\ \text{a Chern–Simons action on an } (n-1)\text{-manifold,} \\ \hat{\lambda} \in \widehat{H}^n(B_\nabla G;\mathbb{Z}) &\ \rightsquigarrow\ \text{an action on an } (n-1)\text{-manifold carrying both at once (§1.4).} \end{aligned} \tag{1}$$

Coefficients and phases. We write $\mathbb{R}/\mathbb{Z}$ throughout and reserve U(1) for quotations from the literature; the two are the same group. Actions are always exponentiated as e^{iS} with S defined modulo 2π, so a pairing valued in $\mathbb{R}/\mathbb{Z}$ enters as $S = 2\pi\langle -,-\rangle$.

Shadow maps. The symbols α and β always denote *maps* out of a set of anomalies, never elements of their targets:

$$\begin{aligned} \{\text{perturbative anomalies}\} &\xrightarrow{\alpha} \mathrm{Free}\big[H^2\big((\Omega^{d-1}G)_0;\mathbb{Z}\big)\big]\,, \\ \{\text{global anomalies}\} &\xrightarrow{\beta} \mathrm{Hom}(\pi_d[G],\mathbb{R}/\mathbb{Z}). \end{aligned} \tag{2}$$

A specific value is written $\alpha(x)$ or $\beta(x)$. The classifying groups proposed below carry maps of their own into the same two targets, the d-fold transgression (15) and eval $= h^*$ of (21); those are never written α or β.

Standing caveat. Every identification below between a set of anomalies and a cohomology group is a *proposal*; none is claimed to be complete or faithful, and the qualification is not repeated. What is established is that each proposed group admits a canonical map to the corresponding target in (2), compatible with α and β.

1 Anomalies and their cohomological proposals

This section is organized around two canonical "shadows" of an anomaly on the spherical spacetime $\mathcal{M} = \mathbb{S}^d$. The sphere arises from flat space: a finite-action gauge field on $\mathbb{R}^d$ is pure gauge at infinity and so extends, topologically, to a bundle over the one-point compactification $\mathbb{S}^d$, and gauge transformations tending to the identity at infinity become based maps $\mathbb{S}^d \to G$. Although $\mathbb{R}^d$ is contractible, the boundary condition makes the sphere, not the plane, carry the topology.

A perturbative anomaly determines the curvature class of the determinant line bundle; when that curvature vanishes, a global anomaly determines its holonomy character under large gauge transformations. These two maps are α and β of (2), and they are intrinsic to the $\mathbb{S}^d$ probe. The cohomological proposals are to be read as candidate universal realizations, one dimension higher, that explain why such canonical maps exist, not as classification theorems.

The objects below sit in the following degrees. By (1) the last two blocks are both $(d+1)$-dimensional responses and both implement inflow; they differ in what they are functions of.

response functional	lives on	class degree	classifying group	boundary anomaly
SPT phase / gauging weight	$\mathcal{M}^d$	d	$H^d(BG;\mathbb{R}/\mathbb{Z})$	—
topological (function of $[\mathcal{P}]$ alone)	$\mathcal{Y}^{d+1}$	$d+1$	$H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$	global
• continuous (θ-angle)			$(\mathbb{R}/\mathbb{Z})^{b_{d+1}}$	
• discrete torsion			$\mathrm{Tor}[H^{d+2}(BG;\mathbb{Z})]$	
Chern–Simons (depends on A)	$\mathcal{Y}^{d+1}$	$d+2$	$\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})]$	perturbative

A group in degree $d+1$ reads both as a $(d+1)$-dimensional SPT phase and as the anomaly of a d-dimensional theory; identifying the two readings is inflow. For example $H^3(\mathbb{Z}_N;\mathbb{R}/\mathbb{Z}) = \mathbb{Z}_N$ is at once the 2+1-dimensional family containing the Levin–Gu model and the anomaly of a 1+1-dimensional theory with an anomalous $\mathbb{Z}_N$, and $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2;\mathbb{R}/\mathbb{Z}) = \mathbb{Z}_2$ is the pair (trivial, cluster) of 1+1-dimensional SPT phases. For the Chern–Simons row, $G = \mathrm{U}(1)$ with $d = 2$ gives $\mathrm{Free}[H^4(B\mathrm{U}(1);\mathbb{Z})] = \mathbb{Z}$, the level of the bosonic integer quantum Hall response ($\sigma_{xy} = 2k$; odd levels require a spin structure and lie outside this ansatz), while $H^3(B\mathrm{U}(1);\mathbb{R}/\mathbb{Z}) = 0$. The two $(d+1)$-dimensional blocks are assembled in §1.4 into a single differential-cohomology group $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z})$, of which the topological block is the flat subgroup, the θ-angles the identity component, and the Chern–Simons row the curvature image.

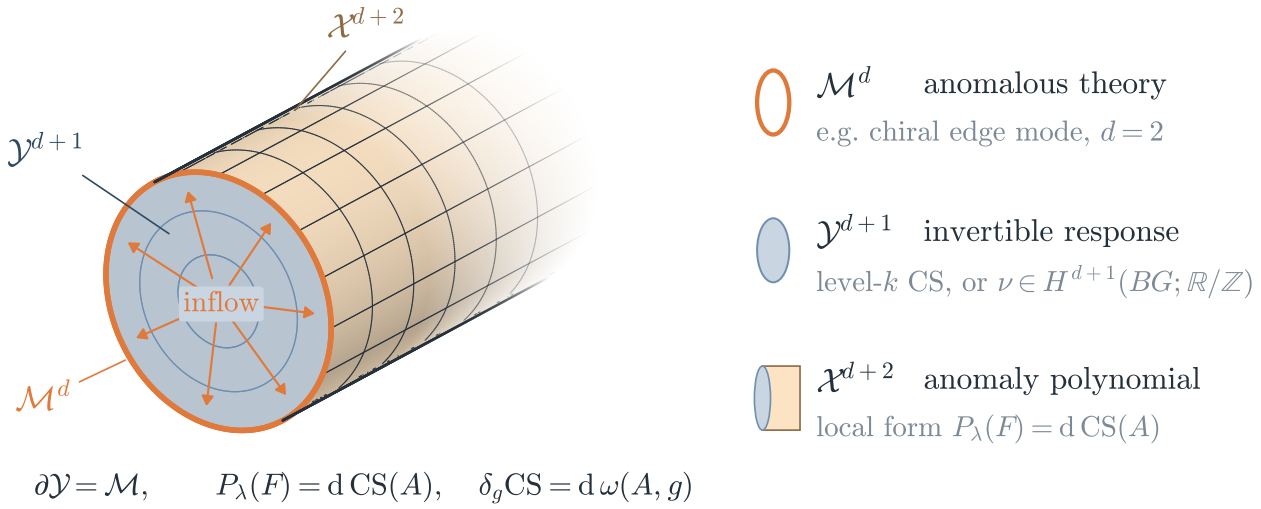


Figure 1: The three dimensions of the Conventions. The anomalous theory lives on $\mathcal{M}^d = \partial\mathcal{Y}$ and the invertible response on $\mathcal{Y}^{d+1}$, which flows into it. One dimension further up sits only a local form: the anomaly polynomial, related to the response by the descent $P_\lambda(F) = \mathrm{d}\ \mathrm{CS}(A)$, $\delta_g\,\mathrm{CS} = \mathrm{d}\,\omega(A,g)$. Since $\partial\partial = \emptyset$, a bounding $\mathcal{X}$ with $\partial\mathcal{X} = \mathcal{Y}$ exists only for closed $\mathcal{Y}$, where it is used to compute $e^{iS_{\mathrm{CS}}}$ through (9) (Fig. 3); the cylinder is drawn so that $\mathcal{X}$ projects onto $\mathcal{Y}$, not as a manifold bounding it.

1.1 The response one dimension up

1.1.1 Anomaly inflow

The link between an anomaly and a $(d+1)$-dimensional action functional is not heuristic. The partition function of an anomalous theory is not a number but a section of a line bundle over the space of backgrounds, equivalently a vector in the state space of a $(d+1)$-dimensional invertible theory [4]. The two shadows of (2) are the curvature and the holonomy of that line bundle: the curvature, computed by the family index theorem for $\mathrm{Det}\,\not{D}$, is the perturbative anomaly, and the holonomy around non-contractible loops carries the global one. The holonomy is a character of π_1, i.e. a value of β, only when the curvature vanishes (§1.3.1). In Fig. 1 the first is read off from the anomaly polynomial $P_\lambda(F)$ on $\mathcal{X}^{d+2}$, and the second from the invertible theory on $\mathcal{Y}^{d+1}$, evaluated on the mapping torus constructed next.

The global shadow has a direct geometric derivation that makes the dimension shift visible. Let $\mathcal{P} \to \mathcal{M}^d$ be a background bundle and $g \in \mathrm{Aut}(\mathcal{P})$. A path of connections from A to A^g is a closed loop in the quotient by gauge transformations, and the same loop is encoded by the gauge mapping-torus bundle

$$\mathcal{P}_g := (\mathcal{P} \times [0,1])/(p,1) \sim (g(p),0) \longrightarrow \mathcal{M}^d \times \mathbb{S}^1. \tag{3}$$

The base is the ordinary product $\mathcal{M}^d \times \mathbb{S}^1$; the twist lies in the bundle fibers, not in a diffeomorphism of spacetime. The anomalous phase is then

$$\exp\big(iW[A^g] - iW[A]\big) = \mathcal{Z}_{\mathrm{anom}}[\mathcal{M}^d \times \mathbb{S}^1, \mathcal{P}_g], \tag{4}$$

where $\mathcal{Z}_{\text{anom}}$ is a $(d+1)$-dimensional invertible response theory [4]. This is the origin of the slogan

$$d\text{-dimensional gauge anomaly} \quad \longleftrightarrow \quad (d+1)\text{-dimensional invertible/SPT response.}$$

When G is a global symmetry probed by a background field, the bulk is an SPT phase in the broad sense of a symmetry-protected invertible phase; §1.2.3 compares this with the narrow, cohomological sense.

Remark 1.1. *Inflow is not restricted to the global sector: a classic example is the four-dimensional chiral anomaly with a five-dimensional Chern–Simons bulk. What depends on the sector is which kind of $(d+1)$-dimensional functional realizes the bulk.*

The question is therefore which functionals of a background G-field are available in dimension $d+1$.

1.1.2 Action functionals: metric-independent, topological, quantized

Definition 1.2. *Chern–Simons functionals are often called topological, but they depend on the connection; they are topological only in the sense that the spacetime metric does not enter. In this note we call such a functional metric-independent, and reserve topological for functionals that are metric-independent and, in addition, invariant under arbitrary deformation of the fields, i.e. locally constant on the space of fields, hence functions of the isomorphism class $[\mathcal{P}]$ alone.*

Definition 1.3. *We call a functional of the gauge field alone quantized if its coefficients must take discrete values to keep the exponentiated action e^{iS} well defined.*

Two mechanisms fall under Definition 1.3. For Chern–Simons the invariant is real-valued with an ambiguity in $2\pi\mathbb{Z}$, and integrality of the coefficient cancels the ambiguity. For the discrete torsion terms below the invariant is already $\mathbb{R}/\mathbb{Z}$-valued, and an element of $\mathbb{R}/\mathbb{Z}$ can be multiplied only by integers.

The two definitions are independent; crossing them gives

	continuous coupling	quantized coupling
topological (function of $[\mathcal{P}]$)	θ-angle $(\mathbb{R}/\mathbb{Z})^{b_n}$	discrete torsion $\mathrm{Tor}[H^{n+1}(BG;\mathbb{Z})]$
metric-indep. non-topological (depends on A)	— *empty: the level is integral*	Chern–Simons $\mathrm{Free}[H^{n+1}(BG;\mathbb{Z})]$

Here n is the dimension of the closed manifold on which the functional is evaluated; $n = d+1$ gives the second and third blocks of the table in §1, and $n = d$ the SPT layer of §2. Fig. 2 draws each cell as its space of couplings.

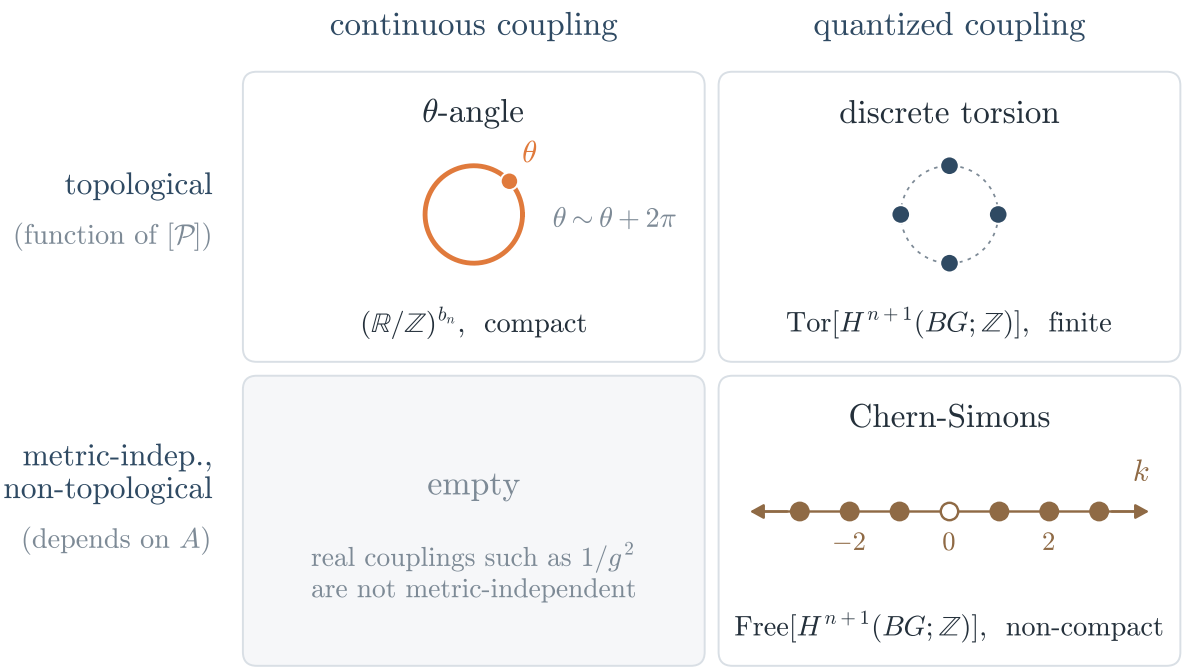


Figure 2: The table above, with each cell drawn as its space of couplings. Topological terms have compact coupling spaces, a circle for each θ-angle and a finite set for discrete torsion, as (7) requires; the Chern–Simons level runs over the non-compact lattice $\mathbb{Z}$. The two quantized cells are quantized for different reasons: an $\mathbb{R}/\mathbb{Z}$-valued pairing admits only integer multiples, whereas for Chern–Simons integrality cancels the $2\pi\mathbb{Z}$ ambiguity of a real-valued invariant.

Topological actions with continuous coupling. For closed $\mathcal{M}$ of dimension n the pairing $Q[\mathcal{P}] = \langle \lambda, [\mathcal{P}] \rangle$ with $\lambda \in \mathrm{Free}[H^n(BG;\mathbb{Z})]$ is a single number, no choice entering its definition, so $e^{i\theta Q}$ makes sense for arbitrary $\theta \in \mathbb{R}$. Integrality of Q is not what permits a continuous coefficient; it makes θ periodic, $e^{i(\theta+2\pi)Q} = e^{i\theta Q}$, so that the parameter space is a circle and not a line. This is the origin of the factor $(\mathbb{R}/\mathbb{Z})^{b_n}$ in (6). The period is fixed by the lattice of values Q attains, in the chosen normalization and on the class of manifolds considered: for $SU(N)/\mathbb{Z}_N$ the instanton number lies in $\frac{1}{N}\mathbb{Z}$ on spin manifolds, and the period of θ is $2\pi N$.

Topological actions with discrete coupling. These are the terms built from a class $\tilde{\lambda} \in H^n(BG;\mathbb{R}/\mathbb{Z})$ that represents a nonzero element of the component group $\pi_0 H^n(BG;\mathbb{R}/\mathbb{Z}) \simeq \mathrm{Tor}[H^{n+1}(BG;\mathbb{Z})]$, evaluated as $S[\mathcal{P}] = 2\pi\langle f_{\mathcal{P}}^* \tilde{\lambda}, [\mathcal{M}] \rangle$. Like the θ-terms they are functions of $[\mathcal{P}]$ alone: they have no local Chern–Simons density and do not shift under large gauge transformations. The available choices form a finite group.

Quantized but non-topological actions. Chern–Simons functionals, constructed in §1.1.3, belong to neither class above. The local expression written in a trivialization shifts under a large gauge transformation, and canceling that shift in e^{iS} forces the level to be integral; once it is, the action itself is invariant. The parameter space is $\mathbb{Z}$, which is not compact, and by Definition 1.2 these functionals are not topological, since they depend on A and not only on $[\mathcal{P}]$.

The lower-left cell of the table is empty because, for compact G, the metric-independent local functionals of A that are not topological are the Chern–Simons functionals, whose level cannot be continuous; continuous couplings such as $1/g^2$ multiply metric-dependent functionals.
A Chern–Simons term cannot be confused with a θ-term, although both are written in terms of $P_\lambda(F)$, because for G compact connected Lie the two never occur in the same spacetime dimension. In dimension n a θ-term needs $\mathrm{Free}[H^n(BG;\mathbb{Z})] \neq 0$ and a Chern–Simons term needs $\mathrm{Free}[H^{n+1}(BG;\mathbb{Z})] \neq 0$; the free part is concentrated in even degrees (Note 1.4), so the former requires n even and the latter n odd. Operationally, *for a θ-term the spacetime is the closed manifold one integrates over, whereas for a Chern–Simons term the spacetime is its boundary.* The same class $\lambda = c_2$ gives $\theta \int \mathrm{tr}\, F \wedge F$ in four dimensions and $k \int \mathrm{tr}(A\,\mathrm{d}A + \frac{2}{3}A^3)$ in three.

The two topological classes together. Within the ordinary-cohomology ansatz the topological response terms in dimension $d+1$, continuous and discrete, are parametrized by

$$H^{d+1}(BG;\mathbb{R}/\mathbb{Z}). \tag{5}$$

The Chern–Simons sector lies outside it, in $\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})]$. Since $\mathbb{R}/\mathbb{Z}$ is divisible and hence injective as a coefficient group, the universal-coefficient decomposition gives

$$\begin{aligned} H^{d+1}(BG;\mathbb{R}/\mathbb{Z}) &\simeq \mathrm{Hom}(H_{d+1}(BG;\mathbb{Z}),\mathbb{R}/\mathbb{Z}) \\ &\simeq \underbrace{(\mathbb{R}/\mathbb{Z})^{b_{d+1}}}_{\text{continuous part}} \oplus \underbrace{\mathrm{Tor}[H^{d+2}(BG;\mathbb{Z})]}_{\text{discrete part}} \end{aligned} \tag{6}$$

The Betti number b_{d+1} counts the copies of $\mathbb{R}/\mathbb{Z}$, whose elements are the distinct θ angles.

Why the continuous couplings are periodic. That continuous topological couplings take values in $\mathbb{R}/\mathbb{Z}$ rather than in $\mathbb{R}$ is a general feature. A topological functional is locally constant on the space of fields, so it is a function of the topological sector. Locality, i.e. multiplicativity under disjoint union together with invariance under bordism, makes e^{iS} a character of a discrete abelian group Γ of sectors: the bordism group of pairs $(\mathcal{M},\mathcal{P})$, replaced in the present ansatz by $H_n(BG;\mathbb{Z})$. The group of topological terms is therefore the Pontryagin dual

$$\{\text{topological terms}\} = \mathrm{Hom}(\Gamma,\mathbb{R}/\mathbb{Z}) = \widehat{\Gamma}, \qquad \Gamma \text{ discrete} \implies \widehat{\Gamma} \text{ compact}. \tag{7}$$

A compact abelian group contains no factor of $\mathbb{R}$: $\Gamma \simeq \mathbb{Z}^b \oplus T$ gives $\widehat{\Gamma} \simeq (\mathbb{R}/\mathbb{Z})^b \times \widehat{T}$, which is (6). Irrational combinations such as $\theta Q_1 + \sqrt{2}\,\theta Q_2$ are not counterexamples: they trace dense lines on the torus $(\mathbb{R}/\mathbb{Z})^2$. Non-compact coupling spaces do exist, $\mathbb{R}_{>0}$ for the Yang–Mills coupling in $\frac{1}{g^2}\int \mathrm{tr}\, F\wedge\star F$ and $\mathbb{Z}$ for the Chern–Simons level, but neither functional is locally constant on the space of fields, so neither falls under (7). The complexified coupling $\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}$ displays both behaviors: its parameter space is a cylinder, compact in the topological direction and not in the other.

Note 1.4.

- (G *compact connected Lie.*) $H^*(BG;\mathbb{R})$ *is a polynomial ring on even-degree generators. Hence for* d *even,* $b_{d+1}=0$ *and* $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})\simeq \mathrm{Tor}[H^{d+2}(BG;\mathbb{Z})]\subset H^{d+2}(BG;\mathbb{Z})$.
- (G *compact connected Lie.*) *For* d *odd,* $H^{d+2}(BG;\mathbb{Z}) = \mathrm{Tor}[H^{d+2}(BG;\mathbb{Z})] \simeq \pi_0 H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$*; in particular* $\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})]=0$.
- (G *finite.*) *Then* $H^n(BG;\mathbb{R}) = 0$ *for all* $n>0$*, hence* $b_n = 0$ *in every degree,* $\mathrm{Free}[H^n(BG;\mathbb{Z})]=0$ *in every degree* $n>0$*, and* $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})\simeq \mathrm{Tor}[H^{d+2}(BG;\mathbb{Z})]$ *for* both *parities of* d.

The finite case is the one relevant to §2. Finiteness of G has three consequences:

- $\mathfrak{g}=0$, so there is no connection form and no curvature: the pair $(\mathcal{P},A)$ of §2.1 collapses to a single datum.
- No continuous couplings survive in (6): there are no θ-angles, only discrete torsion.
- No free part survives either, so every anomaly of a finite group is a torsion class, that is, purely global; this is the cohomological counterpart of the absence of infinitesimal gauge transformations.

In particular the Chern–Simons layer does not merge with the discrete layer for finite G; it is empty.

Terms versus phases. A connected family cannot label distinct phases: θ can be deformed continuously to zero while the theory remains invertible, so all values of θ lie in one phase. The correct statement is

$$\begin{aligned}\{\text{topological terms}\} &= H^n(BG;\mathbb{R}/\mathbb{Z}),\\ \{\text{distinct phases}\} &= \pi_0\big(H^n(BG;\mathbb{R}/\mathbb{Z})\big) = \mathrm{Tor}[H^{n+1}(BG;\mathbb{Z})],\end{aligned} \tag{8}$$

so the discrete-coupling terms are exactly the terms surviving π_0. The distinction has content only when $b_n\neq 0$. For continuous G it means that the map from anomalies to phases in one higher dimension is surjective but not injective, its kernel being the continuous directions. Wen's example [8] is the 3+1-dimensional U(1) term $\int \frac{\theta}{2!(2\pi)^2}\partial_\mu A_\nu \partial_\lambda A_\gamma \epsilon^{\mu\nu\lambda\gamma}$, whose different values of θ give different 2+1-dimensional anomalies but phases connected without transition. The background-field response already separates the values of θ; the gauge field need not be dynamical. For finite G the kernel is trivial by Note 1.4, which is why §2 may use $H^d(BG;\mathbb{R}/\mathbb{Z})$ itself as the classification of SPT phases without passing to π_0. A continuous coupling can nonetheless contribute to a phase classification indirectly: if an additional symmetry acts by $\theta\mapsto-\theta$, only the fixed points $\theta\in\{0,\pi\}$ survive and these are distinct phases; the datum then reappears as a discrete class, an element of π_0, for the enlarged symmetry group (with twisted coefficients if the extra symmetry reverses orientation), consistently with (8).

Why integral cohomology is the natural home. By (1), $H^n(BG;\mathbb{Z})$ is the group of action functionals in spacetime dimension $n-1$ whose coefficients are forced to be discrete. Setting $n = d+2$ gives dimension $d+1$, where the anomaly of a d-dimensional theory is stored (§1.1.1). So $H^{d+2}(BG;\mathbb{Z})$ carries two descriptions at once: the quantized action functionals in dimension $d+1$, and the anomalies of d-dimensional theories that those actions cancel by inflow.

Discreteness is not what makes an anomaly renormalization-group invariant. An anomaly is a response to background fields, and 't Hooft matching demands that such a response agree between the ultraviolet and the infrared whether the parameter labeling it is discrete or continuous; the continuous class above is as rigid as any torsion invariant. What discreteness contributes is a countable label set, hence the possibility of *enumerating* anomalies rather than merely constraining them.

1.1.3 Chern–Simons functionals

Write $n = d+2$ and let $\lambda \in H^n(BG;\mathbb{Z})$ denote the level. The local Chern–Simons density $\mathrm{CS}[A]$ is a $(d+1)$-form built from a choice of local trivialization of $\mathcal{P}$, so on a topologically nontrivial bundle it does not exist as a global differential form on $\mathcal{Y}$. What is well defined is the exponentiated action, obtained by extending $(\mathcal{P}, A)$ over an n-manifold $\mathcal{X}$ with $\partial\mathcal{X} = \mathcal{Y}$:

$$e^{iS_{\mathrm{CS}}[\mathcal{Y};\mathcal{P},A]} \;=\; \exp\left(i\int_{\mathcal{X}} P_\lambda(F)\right), \tag{9}$$

where P_λ is the invariant polynomial representing 2π times the image of λ in de Rham cohomology.

Example 1.5. *Take $d = 2$, so $\mathcal{Y}$ is three-dimensional and $\mathcal{X}$ four-dimensional, and take $G = SU(N)$ with $\lambda = k\,c_2 \in H^4(BSU(N);\mathbb{Z}) = \mathbb{Z}$. Then*

$$P_\lambda(F) = \frac{k}{4\pi}\,\mathrm{tr}(F\wedge F), \qquad \int_{\mathcal{X}} P_\lambda(F) = 2\pi k\int_{\mathcal{X}}\frac{\mathrm{tr}(F\wedge F)}{8\pi^2} = 2\pi k\,\mathbb{Z} \;\; \text{on closed } \mathcal{X},$$

the last integral being the instanton number. On $\mathcal{Y}$ itself $\mathrm{CS}[A] = \frac{k}{4\pi}\,\mathrm{tr}(A\,\mathrm{d}A + \frac{2}{3}A^3)$, whose exterior derivative is $P_\lambda(F)$: the four-dimensional characteristic form is the anomaly polynomial and the three-dimensional Chern–Simons form is the response.

Two extensions differ by a closed $\mathcal{X}'$ (Fig. 3), so independence of the choice requires $\int_{\mathcal{X}'} P_\lambda \in 2\pi\mathbb{Z}$. This is the quantization of the level in the sense of Definition 1.3, and it is the reason the classification is by $H^n(BG;\mathbb{Z})$ and not by $H^n(BG;\mathbb{R})$.

Remark 1.6 (Existence of the bounding manifold)**.** *Formula (9) is a computational device rather than a definition, since a bounding $\mathcal{X}$ exists only if $(\mathcal{Y},\mathcal{P})$ is null-bordant in $\Omega^{SO}_{n-1}(BG)$. For G compact connected Lie and $n-1$ odd, the only case in which the free part of the level can be nonzero (Note 1.4), that group is torsion, so some multiple $N\cdot(\mathcal{Y},\mathcal{P})$ bounds and one may set $S = \frac{1}{N}\int_{\mathcal{X}} P_\lambda$, at the cost of enlarging the ambiguity to $\frac{2\pi}{N}\mathbb{Z}$. The Cheeger–Simons definition below requires no bounding manifold.*

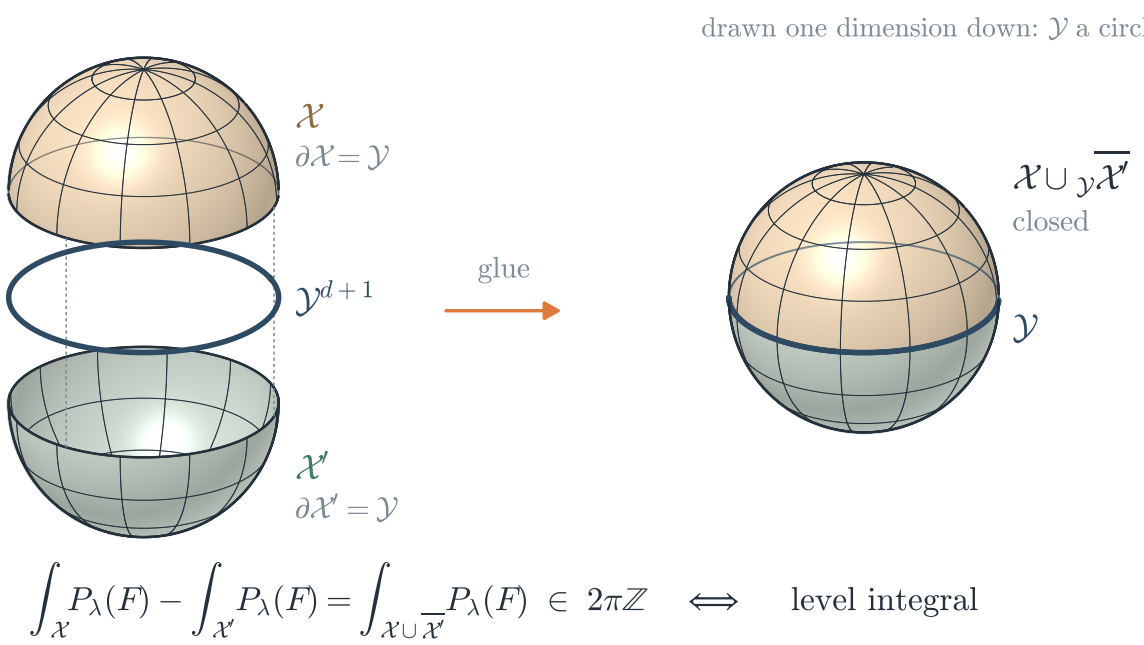


Figure 3: Why the level is quantized. Two fillings $\mathcal{X}$ and $\mathcal{X}'$ of the same closed $\mathcal{Y}$ glue, after reversing the orientation of one, to a closed manifold $\mathcal{X} \cup_{\mathcal{Y}} \overline{\mathcal{X}'}$. The two candidate values of (9) agree precisely when $\int P_\lambda(F) \in 2\pi\mathbb{Z}$ on every closed manifold, i.e. when λ is integral. Drawn one dimension down: $\mathcal{Y}$ is a circle and the fillings are disks.

Invariantly, the level λ lifts to a Cheeger–Simons differential character $\hat{\lambda}$ (§1.4), and the exponentiated action is $e^{iS} = \exp\big(2\pi i \langle \hat{\lambda}(\mathcal{P}, A), [\mathcal{Y}] \rangle\big)$; the dependence on A is carried by $\hat{\lambda}$. The curvature of $\hat{\lambda}$ is $P_\lambda(F)/2\pi$, whose de Rham class is the image of λ under the coefficient map $H^n(BG;\mathbb{Z}) \to H^n(BG;\mathbb{R})$, and the kernel of that map is the torsion subgroup (the coefficient-map lemma, used again in §1.2.1). Hence

$$H^n(BG;\mathbb{Z}) \simeq \underbrace{\mathrm{Free}[H^n(BG;\mathbb{Z})]}_{P_\lambda \neq 0\text{: depends on } A} \oplus \underbrace{\mathrm{Tor}[H^n(BG;\mathbb{Z})]}_{P_\lambda = 0\text{: depends only on } [\mathcal{P}]} . \tag{10}$$

The two summands of (10) are two *kinds of action functional*, not a local and a global piece of one action. For λ torsion the local density vanishes identically, $\hat{\lambda}$ is flat and therefore invariant under deformation of A, and the action depends only on the isomorphism class of $\mathcal{P}$, whether or not A is flat: these are the quantized topological actions with discrete coupling of §1.1.2, and (6) identifies them with the discrete summand of $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$. For finite G they define the topological gauge theories of Dijkgraaf and Witten [3]. The free summand is quantized and metric-independent but not topological, since it depends on the connection.

Caution 1.7. *Two curvatures are in play and must not be conflated. "Flat" in the previous paragraph refers to* $\mathrm{curv}(\hat{\lambda}) = P_\lambda(F)/2\pi$*, not to* F *itself. A torsion level therefore defines a weight in a path integral over* non-flat *bundles, where it appears as a discrete theta angle, constant on each topological sector; the simplest example is two-dimensional* $SO(3)$ *gauge theory weighted by the sign* $(-1)^{\langle w_2(\mathcal{P}), [\mathcal{Y}] \rangle}$*, whose level is the generator of* $H^3(BSO(3);\mathbb{Z}) = \mathbb{Z}_2$*. (The sign* $(-1)^{\mathfrak{P}(w_2)/2}$ *of four-dimensional* $SO(3)$ *theory is not an example:* $H^5(BSO(3);\mathbb{Z}) = 0$*,*

and that sign is the shift $\theta \to \theta + 2\pi$ of the continuous angle.) Conversely, restricting to flat connections does not *kill the free summand. There $P_\lambda(F)$ vanishes pointwise but $\hat\lambda(\mathcal{P}, A)$ need not, and its values are the classical Chern–Simons invariants of flat connections, which depend on the flat structure and not merely on $[\mathcal{P}]$. The reduction to torsion classes used in §2 therefore follows from G being finite, not from a restriction to flat connections. In the language of §1.4,* $\mathrm{curv}(\hat\lambda)$ *is a structure map of $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z})$, whereas F is data of the point of $B_\nabla G$ at which $\hat\lambda$ is evaluated.*

When d is even, (10) is the perturbative/global division of §1.2–§1.3. A perturbative anomaly is a curvature and hence detected only through the coefficient map, so it sees the free summand and nothing else; the torsion summand carries no anomaly polynomial and is invisible to any perturbative expansion. When d is odd the free summand is empty by Note 1.4 and the global side acquires instead the continuous sector $(\mathbb{R}/\mathbb{Z})^{b_{d+1}}$ of (6), which (10) does not see.

1.2 Perturbative anomaly

1.2.1 Gauge fields on spheres

Let $\mathcal{A}$ be the affine space of G-connections on the spacetime $\mathcal{M}$ and $\mathcal{G}$ the group of gauge transformations. The perturbative (local) anomaly of the corresponding gauge theory is the curvature of the determinant line bundle associated to an appropriate Dirac operator over $\mathcal{A}/\mathcal{G}$, and so determines a class

$$\alpha(x) \in \mathrm{im}\big[H^2(\mathcal{A}/\mathcal{G};\mathbb{Z}) \to H^2(\mathcal{A}/\mathcal{G};\mathbb{R})\big]\,. \tag{11}$$

By the coefficient-map lemma of §1.1.3 that image is canonically $\mathrm{Free}[H^2(\mathcal{A}/\mathcal{G};\mathbb{Z})]$, and we use the latter description from here on.

For the spherical spacetime one must distinguish connections on a fixed bundle from the topology of the bundle itself. Let $\mathcal{P}_0 = \mathbb{S}^d \times G$ be the trivial principal bundle and write $\mathcal{A} = \mathcal{A}(\mathcal{P}_0)$ for its affine space of connections. Let $\mathcal{G}_*$ denote the *based* gauge group (gauge transformations restricting to the identity at a basepoint), which unlike $\mathcal{G}$ acts freely. Since $\mathcal{A}$ is affine and hence contractible,

$$\mathcal{A}/\mathcal{G}_* \;\simeq\; B\mathcal{G}_*, \qquad \mathcal{G}_* = \mathrm{Map}_*(\mathcal{M}, G) \overset{\mathcal{M}=\mathbb{S}^d}{=} \Omega^d G, \tag{12}$$

and therefore $\mathcal{A}/\mathcal{G}_* \simeq B\Omega^d G \simeq (\Omega^{d-1}G)_0$, the basepoint component; the component is needed because $B\Omega Y \simeq Y$ holds only for Y connected, and $\Omega^{d-1}G$ need not be.

Fixing $\mathcal{P}_0$ isolates the topology of gauge transformations from that of the background bundle. The two are different homotopy groups: for G connected,

$$\mathrm{Bun}_G(\mathbb{S}^d) = \pi_0[\Omega^{d-1}G] = \pi_{d-1}[G], \qquad \pi_0[\mathcal{G}_*] = \pi_0[\Omega^d G] = \pi_d[G].$$

The first labels G-bundles on $\mathbb{S}^d$ by their clutching classes; the second labels large based gauge transformations within the trivial sector. "Based" means $g(x_0) = e$, not $[g] = 0$. (For G

disconnected the first group must be further quotiented by the conjugation action of $\pi_0[G]$; we assume G connected whenever clutching classes are used.)

In the trivial sector on $\mathcal{M} = \mathbb{S}^d$ the perturbative anomaly thus determines a class in the cohomology of the basepoint component,

$$\{\text{perturbative anomalies}\} \xrightarrow{\alpha} \mathrm{Free}\big[H^2\big((\Omega^{d-1}G)_0;\mathbb{Z}\big)\big]\,, \tag{13}$$

which is the first of the two shadow maps of (2). The cohomological proposal supplies a universal class whose transgression produces this curvature class.

1.2.2 The cohomological proposal

The proposal is that the perturbative anomalies of a d-dimensional theory are captured by

$$\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})], \qquad \text{where} \qquad H^{d+2}(BG;\mathbb{Z}) \simeq \mathcal{H}_B^{d+1}(G;\mathbb{R}/\mathbb{Z}), \tag{14}$$

$\mathcal{H}_B$ standing for Borel group cohomology; the displayed isomorphism is Wigner's theorem, valid for compact Lie G. It is *not* the same statement as (6): for continuous G the two groups $\mathcal{H}_B^{d+1}(G;\mathbb{R}/\mathbb{Z})$ and $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ differ, the former carrying the free part $\mathbb{Z}^{b_{d+2}}$ and the latter the divisible part $(\mathbb{R}/\mathbb{Z})^{b_{d+1}}$; they agree in every degree only for G finite.

The group in (14) classifies the Chern–Simons levels of §1.1.3, hence the terms relevant to the Adler–Bell–Jackiw anomalies, and Note 1.4 makes it nonzero only for d even. It admits a canonical map to the target of (13),

$$\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})] \longrightarrow \mathrm{Free}\big[H^2\big((\Omega^{d-1}G)_0;\mathbb{Z}\big)\big]\,, \tag{15}$$

obtained by iterating the pullback induced by $\mathrm{eval} : \Sigma\Omega X \to X$,

$$\begin{array}{ccccc} [X, K(\mathbb{Z}, m+1)] & \xrightarrow{\mathrm{eval}^*} & [\Sigma\Omega X, K(\mathbb{Z}, m+1)] & = & [\Omega X, K(\mathbb{Z}, m)] \\ \| & & & & \| \\ H^{m+1}(X;\mathbb{Z}) & & & & H^m(\Omega X;\mathbb{Z}) \end{array} \tag{16}$$

Each application lowers the degree by one and loops the space once, so starting from $X = BG$ and $m + 1 = d + 2$ and using $\Omega BG \simeq G$ one obtains after k steps $H^{d+2-k}(\Omega^{k-1}G;\mathbb{Z})$. Taking $k = d$ lands in $H^2(\Omega^{d-1}G;\mathbb{Z})$; restriction to the basepoint component and passage to the free quotient, which are separate final steps, then give (15) (Fig. 4).

1.2.3 The bulk of a perturbative anomaly lies outside $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$

By Remark 1.1 the perturbative anomaly has a bulk like any other, but that bulk is *not* an SPT phase in the narrow sense of a class in $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$, for three reasons internal to this note.

First, the bulk is $\exp(ik\int_{\mathcal{Y}}\mathrm{CS}(A))$, which depends on A and not only on $[\mathcal{P}]$: by Definition 1.2 it is metric-independent but not topological, whereas every element of $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ is a function of $[\mathcal{P}]$ alone. Second, the level lives in $\mathrm{Free}[H^{d+2}(BG;\mathbb{Z})]$, and (6) shows that the

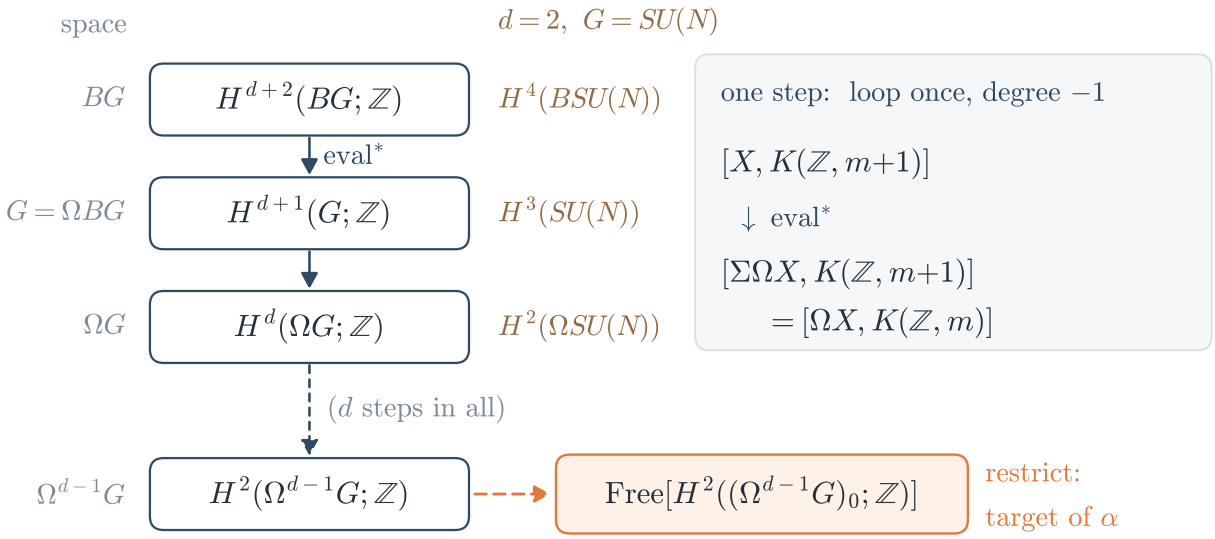


Figure 4: The transgression (15) as a ladder. Each step is the pullback along eval : $\Sigma\Omega X \to X$ (boxed on the right), which loops the space once and lowers the degree by one; after d steps one lands in $H^2(\Omega^{d-1}G;\mathbb{Z})$. The restriction to the basepoint component (dashed, orange) is a separate step and lands in the target of α. The brown column is the case $d=2$, $G=SU(N)$, where the ladder has two steps.

free part is absent from $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$. It is present in the Borel group $\mathcal{H}_B^{d+1}(G;\mathbb{R}/\mathbb{Z})$ of (14), so whether such a phase counts as "group-cohomology" depends on which of the two groups is meant. Third, for G finite the free summand is empty altogether by Note 1.4, so neither the perturbative anomaly nor its bulk exists; this is why §2 never meets this layer.

In the broad sense of any invertible theory with symmetry, the Chern–Simons bulk does qualify as an SPT phase. A concrete contrast: a two-dimensional chiral theory with U(1) symmetry has a perturbative anomaly whose bulk is three-dimensional U(1) Chern–Simons, a quantized Hall response; here $H^3(B\mathrm{U}(1);\mathbb{R}/\mathbb{Z})=0$ while the level lives in $H^4(B\mathrm{U}(1);\mathbb{Z})=\mathbb{Z}$. By contrast a two-dimensional theory with an anomalous $\mathbb{Z}_2$, classified by $H^3(\mathbb{Z}_2;\mathbb{R}/\mathbb{Z})=\mathbb{Z}_2$, has the Levin–Gu model as its bulk, a topological response in the narrow sense.

1.3 Global anomaly

1.3.1 Holonomy of the determinant line

Assume first that the perturbative anomaly has vanished, so the determinant line bundle is flat. It can still be nontrivial as a flat bundle, and its holonomy around loops in the space of backgrounds is the global anomaly: a rank-one local system, i.e. an element of $\mathrm{Loc}_{\mathbb{R}/\mathbb{Z}}(Z) = \mathrm{Hom}(\pi_1[Z],\mathbb{R}/\mathbb{Z})$ for $Z=\mathcal{A}/\mathcal{G}_*$. In the fixed trivial sector on $\mathcal{M}=\mathbb{S}^d$, the fibration $\mathcal{G}_* \to \mathcal{A} \to \mathcal{A}/\mathcal{G}_*$ with $\mathcal{A}$ contractible gives

$$\pi_1[\mathcal{A}/\mathcal{G}_*] \simeq \pi_0[\mathcal{G}_*] = \pi_d[G],$$

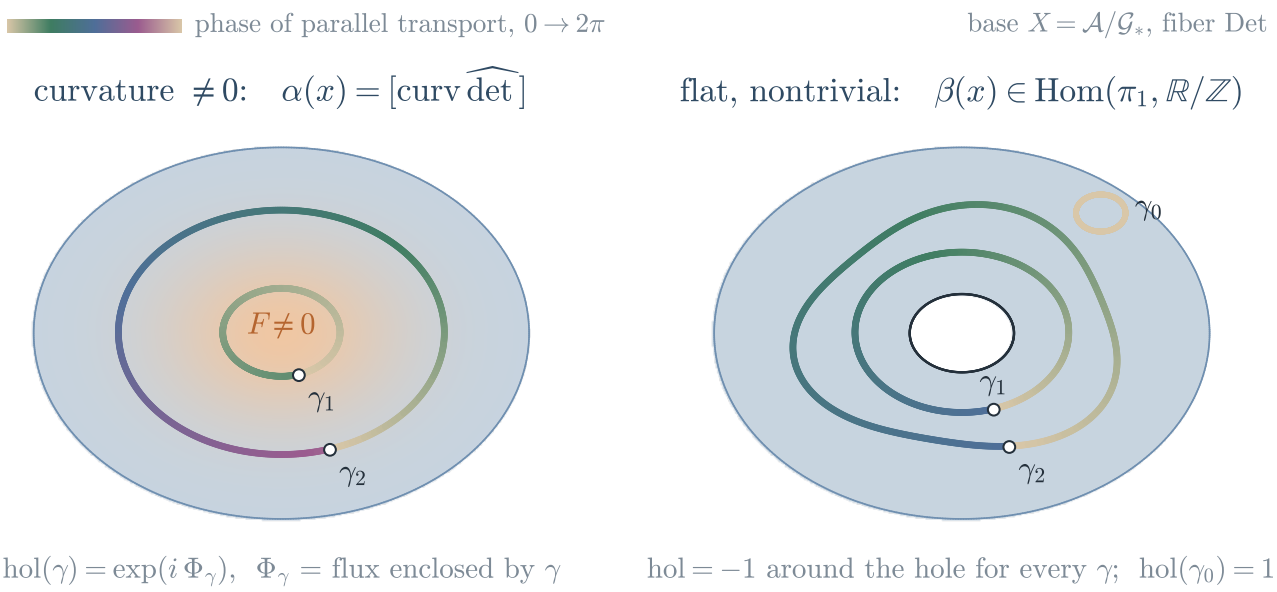


Figure 5: The two shadows as parallel transport in the determinant line over $\mathcal{A}/\mathcal{G}_*$; color is the accumulated phase. Left: with curvature, the holonomy of a contractible loop is the flux it encloses, and grows with the loop; this curvature is what α records. Right: with vanishing curvature, contractible loops (γ_0) have trivial holonomy, and every loop around the hole returns with the same phase, here -1, however it is deformed; this homotopy-invariant holonomy is what β records. The right picture exists only once the left one is flat; §1.4.2 identifies the two as the curvature and flat part of one class (Fig. 8).

so the holonomy is the second shadow map of (2),

$$\{\text{global anomalies}\} \xrightarrow{\beta} \operatorname{Loc}_{\mathbb{R}/\mathbb{Z}}(\mathcal{A}/\mathcal{G}_*) = \operatorname{Hom}(\pi_d[G], \mathbb{R}/\mathbb{Z}). \tag{17}$$

The vanishing of the perturbative anomaly is what makes β well defined (Fig. 5). This ordering, α first and β only on its kernel, is not a convention: §1.4.2 realizes both shadows as structure maps of a single class in $\widehat{H}^2(\mathcal{A}/\mathcal{G}_*; \mathbb{Z})$, which exists whether or not the perturbative anomaly vanishes.

1.3.2 The ordinary-cohomology realization

The mapping-torus argument of §1.1.1 says only that the anomaly is stored one dimension higher. To obtain a universal response defined on arbitrary closed $(d+1)$-manifolds $\mathcal{Y}$ and arbitrary G-bundles $\mathcal{Q} \to \mathcal{Y}$, the ordinary-cohomology proposal chooses

$$\nu \in H^{d+1}(BG; \mathbb{R}/\mathbb{Z}), \qquad \mathcal{Z}_\nu[\mathcal{Y}, \mathcal{Q}] = \exp\big(2\pi i \,\langle f_{\mathcal{Q}}^* \nu, [\mathcal{Y}]\rangle\big)\,,$$

with $f_{\mathcal{Q}} : \mathcal{Y} \to BG$ the classifying map. Its restriction to (3) gives

$$\exp\big(iW[A^g] - iW[A]\big) = \mathcal{Z}_\nu[\mathcal{M}^d \times \mathbb{S}^1, \mathcal{P}_g].$$

Thus (5) is an ordinary-cohomology realization of the one-higher-dimensional response, not a definition of every anomaly.

The trivial based spherical sector. Now specialize to $\mathcal{M}^d = \mathbb{S}^d$, the trivial bundle $\mathcal{P}_0 = \mathbb{S}^d \times G$, and a based gauge transformation $g(x_0) = e$; as in §1.2.1, this isolates the topology of g from that of the bundle. Every slice $\mathbb{S}^d \times \{t\}$ of $\mathcal{P}_g \to \mathbb{S}^d \times \mathbb{S}^1$ carries the trivial bundle, while $\mathcal{P}_g$ itself can be nontrivial because of the gluing by g.

The two assumptions have separate roles. Triviality of $\mathcal{P}_0$ supplies a trivialization on $\mathbb{S}^d \times \{t_0\}$, while the based condition supplies one on $\{x_0\} \times \mathbb{S}^1$. Hence a classifying map $F_g : \mathbb{S}^d \times \mathbb{S}^1 \to BG$ may be chosen constant on their union

$$\mathbb{S}^d \times \{t_0\} \;\cup\; \{x_0\} \times \mathbb{S}^1 = \mathbb{S}^d \vee \mathbb{S}^1,$$

and therefore factors as

$$\mathbb{S}^d \times \mathbb{S}^1 \xrightarrow{\;q\;} \mathbb{S}^d \wedge \mathbb{S}^1 \simeq \mathbb{S}^{d+1} \xrightarrow{\;\widehat{g}\;} BG. \tag{18}$$

The mapping-torus bundle over $\mathbb{S}^d \times \mathbb{S}^1$ and the clutching bundle over $\mathbb{S}^{d+1}$ are different bundles, their bases being different (Fig. 6); (18) says that they carry the same homotopy datum $[\widehat{g}] \in \pi_{d+1}(BG) \simeq \pi_d[G]$. Consequently the two evaluations agree,

$$\left\langle F_g^*\nu, [\mathbb{S}^d \times \mathbb{S}^1] \right\rangle = \left\langle \widehat{g}^*\nu, [\mathbb{S}^{d+1}] \right\rangle, \tag{19}$$

and the cohomological response produces a spherical character

$$\mathrm{eval}(\nu) : [g] \longmapsto \left\langle \widehat{g}^*\nu, [\mathbb{S}^{d+1}] \right\rangle \in \mathbb{R}/\mathbb{Z}. \tag{20}$$

This is a probe: general spacetimes, nontrivial background bundles, and closed $(d+1)$-manifolds that are not such mapping tori can carry anomaly information invisible in the trivial based spherical sector.

1.3.3 The spherical restriction and the Hurewicz map

The construction above gives the canonical homomorphism

$$\mathrm{eval} : H^{d+1}(BG; \mathbb{R}/\mathbb{Z}) \longrightarrow \mathrm{Hom}(\pi_d[G], \mathbb{R}/\mathbb{Z}), \tag{21}$$

a map out of the *classifying group*, to be compared with the shadow map β of (17) out of the set of anomalies. The universal coefficient isomorphism of (6), the loop–suspension identification $\pi_d[G] \simeq \pi_{d+1}(BG)$ and the Hurewicz map $h : \pi_{d+1}(BG) \to H_{d+1}(BG; \mathbb{Z})$ give the factorization

$$H^{d+1}(BG; \mathbb{R}/\mathbb{Z}) \simeq \mathrm{Hom}(H_{d+1}(BG), \mathbb{R}/\mathbb{Z}) \xrightarrow{\;h^*\;} \mathrm{Hom}(\pi_{d+1}(BG), \mathbb{R}/\mathbb{Z}) \simeq \mathrm{Hom}(\pi_d[G], \mathbb{R}/\mathbb{Z}). \tag{22}$$

Thus $\mathrm{eval} = h^*$: ordinary cohomology sees a spherical large gauge transformation only through its Hurewicz image in homology. Two losses of information follow (Fig. 7).

- $\ker(\mathrm{eval})$ consists of cohomological responses that are invisible to the spherical large-gauge probe; the probe is too weak.

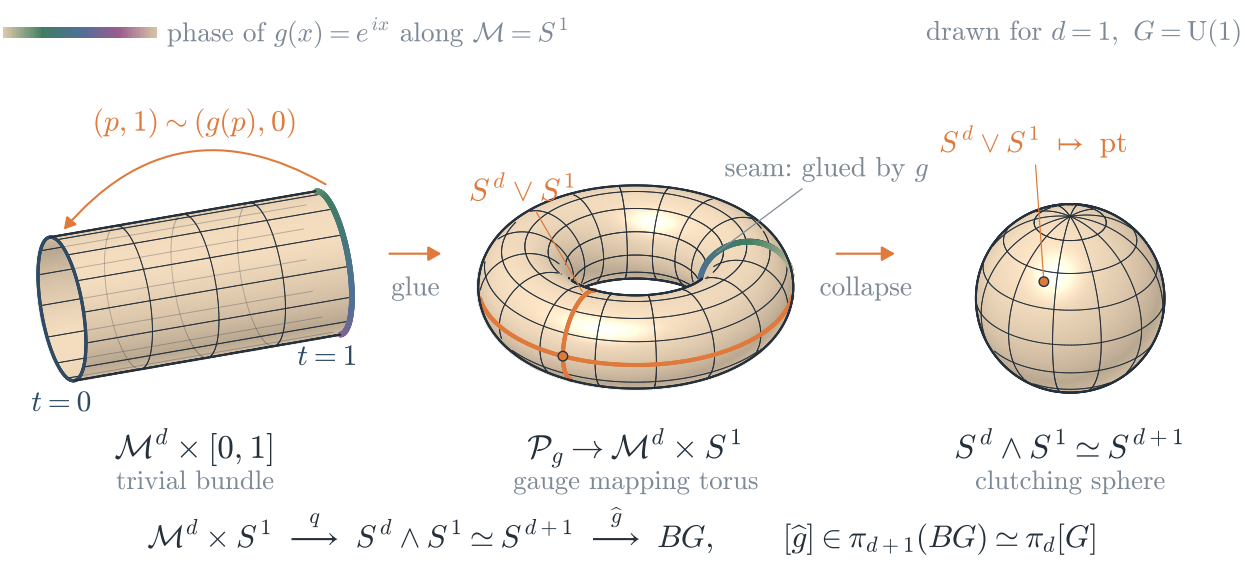


Figure 6: The gauge mapping torus (3) and the collapse (18), drawn for $d = 1$, $G = \mathrm{U}(1)$, $g(x) = e^{ix}$. The base of $\mathcal{P}_g$ is the ordinary product; the twist sits only on the seam, where fibers are glued by g (color = phase of g). Triviality of $\mathcal{P}_0$ and the based condition trivialize $\mathcal{P}_g$ on $\mathbb{S}^d \vee \mathbb{S}^1$ (orange), which is collapsed to a point.

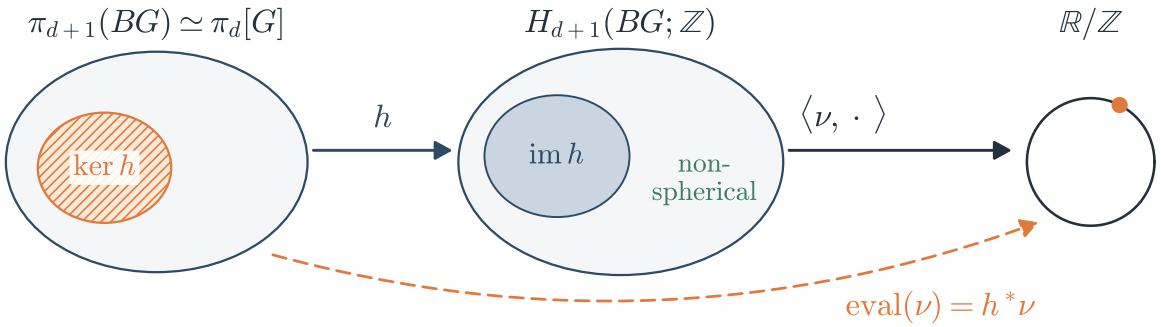
- coker(eval) consists of spherical characters that do not factor through h and hence admit no ordinary-cohomological extension; the classifying group is too small.

Both are examined in §1.4.4.

Remark 1.8 (The spherical probe is empty for finite G)**.** *If G is finite, then $\pi_d[G] = 0$ for every $d \geq 1$: the trivial bundle on $\mathbb{S}^d$ has no nontrivial based gauge transformations,* $\mathrm{Hom}(\pi_d[G], \mathbb{R}/\mathbb{Z}) = 0$*, and the map (21) vanishes. The one-higher-dimensional response need not: for example $H^3(B\mathbb{Z}_2; \mathbb{R}/\mathbb{Z}) = \mathbb{Z}_2$. Such classes must be detected on more general backgrounds, such as lens spaces, rather than on the spherical mapping tori constructed from maps $\mathbb{S}^d \to G$. Finite groups are thus the extreme case of* $\ker(\mathrm{eval})$*.*

1.4 The differential-cohomology synthesis

§1.1.2 sorted the $(d + 1)$-dimensional functionals into three kinds, and §1.2–§1.3 attached a shadow map to each side of the perturbative–global division; every step produced a group of its own. All of them are facets of a single group, the differential cohomology $\widehat{H}^{d+2}(B_\nabla G; \mathbb{Z})$, and the two shadow maps of (2) are the two structure maps of the same construction taken in degree two. The assembly is internal to the ordinary-cohomology ansatz and repairs none of its limitations (§1.4.4).



ker(eval): the probe is too weak

ν lives on cycles outside $\operatorname{im} h$

$G = \mathbb{Z}_2$, $d = 2$: $\pi_2[\mathbb{Z}_2] = 0$,

but $H^3(B\mathbb{Z}_2; \mathbb{R}/\mathbb{Z}) = \mathbb{Z}_2$

coker(eval): the group is too small

characters nonzero on $\ker h$ have no ν

$G = SU(2)$, $d = 4$: $\pi_5(BSU(2)) = \mathbb{Z}_2$,

but $H_5(BSU(2); \mathbb{Z}) = 0$ (Witten)

Figure 7: eval $= h^*$: a cohomological response ν is seen by the spherical probe only through the Hurewicz image. Classes supported on non-spherical cycles are invisible to it (ker eval; Remark 1.8), and characters that are nonzero on $\ker h$ are reproduced by no ν at all (coker eval; (34)). The first is a defect of the probe, the second of the classifying group.

1.4.1 Differential characters

Let X be a smooth manifold. A *differential character* of degree k on X [2] is a homomorphism on smooth $(k-1)$-cycles,

$$\chi : Z_{k-1}(X) \longrightarrow \mathbb{R}/\mathbb{Z}, \qquad \chi(\partial c) = \int_c \omega \mod \mathbb{Z} \quad \text{for every smooth } k\text{-chain } c, \tag{23}$$

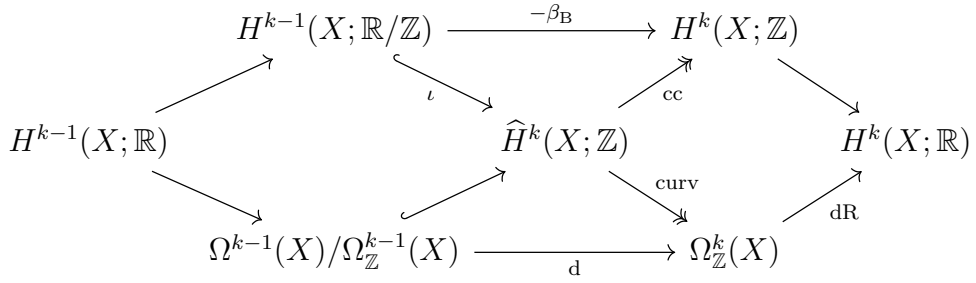
for some k-form ω. The form is determined by χ, closed, and has integral periods; write $\omega = \operatorname{curv}(\chi)$, let $\Omega^k_{\mathbb{Z}}(X)$ denote the group of such forms, and let $\widehat{H}^k(X;\mathbb{Z})$ denote the group of characters. Every character also determines an integral class $\operatorname{cc}(\chi) \in H^k(X;\mathbb{Z})$, whose image under $H^k(X;\mathbb{Z}) \to H^k(X;\mathbb{R})$ is $[\operatorname{curv}(\chi)]_{\mathrm{dR}}$. The group sits in two exact sequences,

$$0 \longrightarrow H^{k-1}(X;\mathbb{R}/\mathbb{Z}) \xrightarrow{\iota} \widehat{H}^k(X;\mathbb{Z}) \xrightarrow{\operatorname{curv}} \Omega^k_{\mathbb{Z}}(X) \longrightarrow 0, \tag{24}$$

$$0 \longrightarrow \Omega^{k-1}(X)/\Omega^{k-1}_{\mathbb{Z}}(X) \longrightarrow \widehat{H}^k(X;\mathbb{Z}) \xrightarrow{\operatorname{cc}} H^k(X;\mathbb{Z}) \longrightarrow 0. \tag{25}$$

The kernel of curv consists of the *flat* characters, which factor through homology; (24) identifies them with $H^{k-1}(X;\mathbb{R}/\mathbb{Z})$, and by Definition 1.2 they are exactly the topological functionals.

Both sequences, together with their compatibilities, are packaged in the character diagram



(26)

in which the two diagonals through the center are (24) and (25), the upper and lower rows close up with the side maps into the Bockstein and de Rham long exact sequences, and every triangle and square commutes. In particular the top triangle reads

$$\mathrm{cc} \circ \iota \;=\; -\beta_{\mathrm{B}}, \tag{27}$$

β_{B} being the Bockstein of $0 \to \mathbb{Z} \to \mathbb{R} \to \mathbb{R}/\mathbb{Z} \to 0$; the sign is a matter of convention and nothing below depends on it. The Bockstein of the Conventions is thus the top edge of (26).

Remark 1.9 (Component group, identity component, and non-canonicity)**.** *As a topological group, $\widehat{H}^k(X;\mathbb{Z})$ has component group $H^k(X;\mathbb{Z})$, realized by* cc*; the topologically trivial characters form the identity component, which contains the torus $H^{k-1}(X;\mathbb{R})/\operatorname{im}\big[H^{k-1}(X;\mathbb{Z})\big] \simeq (\mathbb{R}/\mathbb{Z})^{b_{k-1}}$. Neither (24) nor (25) splits canonically: the canonical data are the three maps ι,* curv*,* cc*, and any direct-sum presentation of $\widehat{H}^k$ or of its subquotients requires choices.*

Remark 1.10 (Meaning of $\widehat{H}^*(B_\nabla G;\mathbb{Z})$)**.** *$BG$ is not a manifold, so the symbol requires a model, and three equivalent routes are available: universal characters in the original sense of [2], i.e. assignments $(\mathcal{Y};\mathcal{P},A) \mapsto \chi(\mathcal{P},A)$ natural under pullback; the differential function spectra of [6]; and the sheaf $B_\nabla G$ on the site of manifolds constructed in [5], the same object that appears in (36). The last route computes the forms: $\Omega^j(B_\nabla G) \simeq (\mathrm{Sym}^{j/2}\,\mathfrak{g}^*)^G$, the invariant polynomials, concentrated in even degrees and all closed. This computation gives (24) and (25) their concrete form in §1.4.3.*

1.4.2 Degree two: the two shadows are one class

For $k = 2$ the group is classical: $\widehat{H}^2(X;\mathbb{Z})$ is the group of isomorphism classes of hermitian line bundles with unitary connection on X, a character being the holonomy of the connection around cycles. Under this identification curv is the curvature two-form, cc the first Chern class, and the flat subgroup $H^1(X;\mathbb{R}/\mathbb{Z}) = \mathrm{Hom}(\pi_1[X],\mathbb{R}/\mathbb{Z}) = \mathrm{Loc}_{\mathbb{R}/\mathbb{Z}}(X)$ is the group of flat line bundles, recorded by their holonomy characters.

Now take $X = \mathcal{A}/\mathcal{G}_*$ and the determinant line bundle with its Bismut–Freed connection [1], giving a class

$$\widehat{\det} \;\in\; \widehat{H}^2(\mathcal{A}/\mathcal{G}_*;\mathbb{Z}). \tag{28}$$

The two shadow maps of (2) are the two structure maps of this one class (Fig. 8). The value of α in (11) is the de Rham class of $\mathrm{curv}(\widehat{\det})$. When that curvature vanishes, $\widehat{\det}$ lies, by

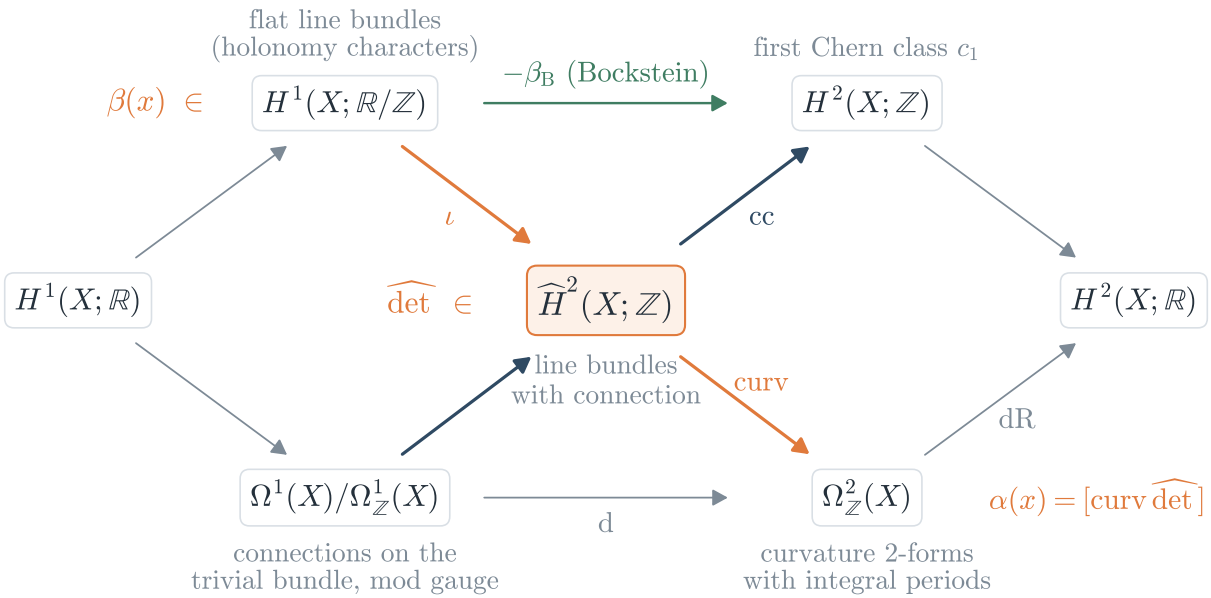


Figure 8: The character diagram (26) in degree two on $X = \mathcal{A}/\mathcal{G}_*$, each vertex named by its line-bundle meaning. The orange diagonal is (24), so β is defined only on $\ker \mathrm{curv}$, and the navy one is (25). For $\widehat{\det}$ of (28), α reads off the curvature and, once that vanishes, β reads off the flat class in $H^1(X;\mathbb{R}/\mathbb{Z})$; the Bockstein triangle (27) gives its torsion Chern class.

exactness of (24), in the flat subgroup $\mathrm{Loc}_{\mathbb{R}/\mathbb{Z}}(\mathcal{A}/\mathcal{G}_*)$, where it *is* the local system of §1.3.1, i.e. the value of β. Since (24) does not split canonically, the flat part of a class, and with it β, is defined only on the kernel of curv; when the perturbative anomaly is nonzero the global information is still carried by $\widehat{\det}$, just not as a separate character of π_1.

A flat determinant line has torsion Chern class, and (27) computes that class as (minus) the Bockstein of the holonomy character; the free part of $\mathrm{cc}(\widehat{\det})$, by contrast, is what α sees through the coefficient map.

1.4.3 Degree $d+2$: the responses assemble

Taking $k = d+2$ on $B_\nabla G$ collects the $(d+1)$-dimensional response functionals of the ordinary ansatz, topological terms and Chern–Simons functionals together, into the single group

$$\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}), \qquad e^{iS[\mathcal{Y};\mathcal{P},A]} = e^{2\pi i\langle \hat{\lambda}(\mathcal{P},A),[\mathcal{Y}]\rangle}, \tag{29}$$

the evaluation being the one already used for Chern–Simons functionals in §1.1.3; it requires no bounding manifold (Remark 1.6). By Remark 1.10 the forms on $B_\nabla G$ are the invariant polynomials, so for G compact the two exact sequences specialize to

$$0 \longrightarrow H^{d+1}(BG;\mathbb{R}/\mathbb{Z}) \xrightarrow{\ \iota\ } \widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}) \xrightarrow{\ \mathrm{curv}\ } \left(\mathrm{Sym}^{\frac{d+2}{2}}\mathfrak{g}^*\right)^G_{\mathbb{Z}} \longrightarrow 0, \tag{30}$$

$$0 \longrightarrow (\mathbb{R}/\mathbb{Z})^{b_{d+1}} \longrightarrow \widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}) \xrightarrow{\ \mathrm{cc}\ } H^{d+2}(BG;\mathbb{Z}) \longrightarrow 0, \tag{31}$$

where the curvature target is the lattice of invariant polynomials with integral periods (zero unless d is even), and the left-hand group of (31) is

$$\Omega^{d+1}(B_\nabla G)/\Omega^{d+1}_{\mathbb{Z}} \simeq H^{d+1}(BG;\mathbb{R})/\operatorname{im}[H^{d+1}(BG;\mathbb{Z})]$$

(zero unless d is odd). Everything in the table of §1 is now a facet of this one group.

- *Component group.* By Remark 1.9 it is $H^{d+2}(BG;\mathbb{Z})$: the quantized labels, in the sense of Definition 1.3. Its free part, the Chern–Simons levels, is detected by curv, which by (10) annihilates exactly the torsion; its torsion part lifts to flat characters, the discrete torsion terms.

- *Identity component.* The torus $(\mathbb{R}/\mathbb{Z})^{b_{d+1}}$ of (31): the θ-angles, continuously connected to zero as characters although nontrivial as functionals.

- *Flat subgroup.* $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ of (30): the topological responses (5). Its internal structure is the extension

$$0 \longrightarrow (\mathbb{R}/\mathbb{Z})^{b_{d+1}} \longrightarrow H^{d+1}(BG;\mathbb{R}/\mathbb{Z}) \xrightarrow{\text{cc}\circ\iota} \operatorname{Tor}[H^{d+2}(BG;\mathbb{Z})] \longrightarrow 0, \tag{32}$$

  which is (6), with (27) identifying the projection as the Bockstein; that (6) is written as a direct sum reflects a choice of splitting, not a canonical one (Remark 1.9).

Note 1.4 is recovered by reading off which outer group survives. For G compact connected Lie and d even the identity component is absent and $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}) \simeq H^{d+2}(BG;\mathbb{Z})$ is discrete: Chern–Simons levels and discrete torsion, nothing continuous. For d odd the curvature target is absent, every class is flat, and $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}) \simeq H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$: θ-angles and discrete torsion, nothing perturbative. For G finite both form groups vanish in every degree and

$$\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z}) \;\simeq\; H^{d+1}(BG;\mathbb{R}/\mathbb{Z}) \;\simeq\; \operatorname{Tor}[H^{d+2}(BG;\mathbb{Z})], \tag{33}$$

which is the situation of §2: the Dijkgraaf–Witten labels exhaust the differential cohomology, all of it flat.

1.4.4 The limits of the assembled ansatz

The flat sector of $\widehat{H}^{d+2}(B_\nabla G;\mathbb{Z})$ is exactly $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$, so both losses of information identified in §1.3.3 survive the assembly. The spherical probe still factors through the Hurewicz map: ker(eval) is unchanged, with Remark 1.8 the extreme case. So is coker(eval), which is the sharper defect, since it concerns the classifying group rather than the probe. The four-dimensional $SU(2)$ anomaly is the standard illustration:

$$\pi_4(SU(2)) \simeq \pi_5(BSU(2)) \simeq \mathbb{Z}_2, \qquad H_5(BSU(2);\mathbb{Z}) = 0, \qquad H^5(BSU(2);\mathbb{R}/\mathbb{Z}) = 0, \tag{34}$$

so a nontrivial spherical homotopy class lies in the kernel of h and no ordinary class, differentially refined or not, can supply its phase.

One repair from inside the cochain framework has been proposed. Wen [8] replaces the flat sector by a homotopy-sensitive group $\mathscr{H}^{d+1}_{\pi}(BG;\mathbb{R}/\mathbb{Z})$, defined by cochains required to be invariant only under homotopy of the classifying map rather than under bounding; since homotopic implies bounding, the ordinary cocycle condition is the stronger one, and the ordinary group $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ is contained in $\mathscr{H}^{d+1}_{\pi}(BG;\mathbb{R}/\mathbb{Z})$. Three qualifications, all of them Wen's own, limit its bearing on these notes: it is not known whether the enlargement is ever strictly larger, even for continuous G; for G finite the two groups coincide, because $\pi_n(BG) = 0$ for $n > 1$ makes a π-cocycle a group cocycle, so the enlargement is invisible to §2; and whether it captures the $SU(2)$ anomaly of (34) is unsettled, that anomaly being fermionic and described by a mod-2 index. Moreover the enlargement acts on the flat sector alone, leaving the curvature image of (30) untouched. What must change is the coefficient theory, not the cochain condition imposed on it.

Invertible field theories are classified by bordism [4, 10], where the role of $H^{d+2}(BG;\mathbb{Z})$ is played by the Anderson dual of a bordism spectrum; this is the framework in which the mod-2 index describing (34) is at home. The construction that does for that classification what differential characters do for ordinary cohomology, namely one object carrying deformation classes, continuous θ-like directions and anomaly polynomials, is the differential model of the Anderson dual of [9]. Pursuing the present notes to that level is deferred.

2 SPT, gauge theory, and cohomology

Scope. Everything below assumes G finite. The simplicial construction imposes two conditions, only one of which concerns flatness.

- **(A) The cocycle condition** $g_{ij}g_{jk} = g_{ik}$ on every 2-simplex. This is what makes the data define a bundle *on* $\mathcal{M}$ at all; every bundle satisfies it, flat or not.
- **(B) Local constancy** of the transition functions, as opposed to smoothness. *This* is flatness, and it amounts to using the constant sheaf $\underline{G}$ rather than $\underline{C^{\infty}(-,G)}$.

For finite G condition (B) is vacuous, since $\mathfrak{g} = 0$ leaves no connection form, but (A) is not. On an $L \times L$ periodic square lattice with $G = \mathbb{Z}_2$, imposing (A) leaves $|H^1(T^2;\mathbb{Z}_2)| = 4$ gauge classes, whereas without it there are $2^{|E|-|V|+1}$, i.e. 32, 1024, 131072 for $L = 2, 3, 4$ (Fig. 9).

Relaxing (A) does not produce non-flat bundles, of which there are none for finite G, but flat bundles on $\mathcal{M}$ with codimension-two *defects* inserted, the plaquette holonomy measuring the flux through each. This is lattice gauge theory with a finite gauge group; in the toric code it is the violation of $B_p = \prod_{e\in\partial p}\sigma^z$, i.e. an m anyon. Changing (B) instead leads out of the combinatorial setting, to the differential refinement $B_{\nabla}G$ of (36) and the Cheeger–Simons characters of §1.1.3 and §1.4.

Finiteness also collapses distinctions that §1.1.2 and §1.4.3 keep apart. With $b_n = 0$ in every degree, the topological terms of a given degree, the SPT phases they label, and the admissible

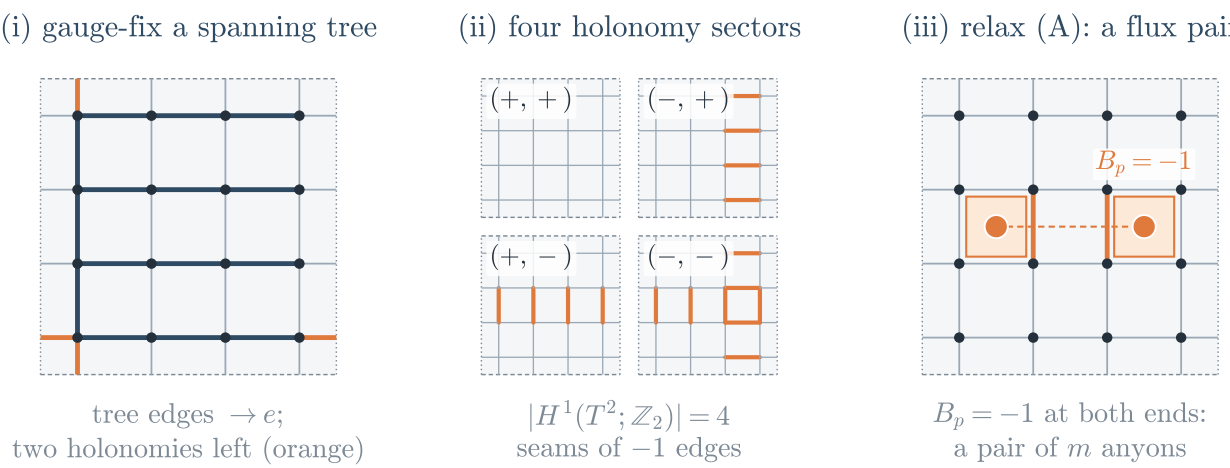


Figure 9: A $\mathbb{Z}_2$ gauge field on the periodic 4×4 square lattice (dashed sides identified). (i) Gauge-fixing a spanning tree (navy) to e leaves the values on the remaining edges fixed by (A) in terms of the two holonomies (orange), cf. (40). (ii) The four sectors of $H^1(T^2; \mathbb{Z}_2)$, each a seam of -1 edges across a non-contractible cycle; the plaquette where two seams cross carries four -1's and is flat. (iii) Dropping (A): a string of -1 edges ending inside the lattice has $B_p = -1$ at both ends, a pair of m anyons.

Dijkgraaf–Witten actions are one and the same set,

$$\{\text{topological terms}\} = \{\text{SPT phases}\} = \{DW \text{ labels}\} = \mathrm{Tor}[H^{d+1}(BG; \mathbb{Z})], \tag{35}$$

which for finite G is all of $H^{d+1}(BG; \mathbb{Z}) \simeq H^d(BG; \mathbb{R}/\mathbb{Z})$. The gap between terms and phases recorded in (8) is empty here, and no continuous couplings appear. Gauging does not enlarge the label set, the class ω being carried across unchanged; what the Dijkgraaf–Witten construction adds is a realization of those labels as topological gauge theories (§2.3).

2.1 Simplicial model for gauge theory

Let $\mathcal{M}$ be the spacetime manifold. The partition function of a pure gauge theory with gauge group G takes the schematic form

$$\mathcal{Z} \sim \sum_{(\mathcal{P}, A)} e^{iS[\mathcal{P}; A]} \tag{36}$$

where the sum runs over the pairs of a principal G-bundle $\mathcal{P}$ over $\mathcal{M}$ and a G-connection A on $\mathcal{P}$. Equivalently, it is a sum over the mapping space $\mathrm{Map}[\mathcal{M}, B_\nabla G]$ where $E_\nabla G \to B_\nabla G$ is the differential refinement of the universal bundle.

At weak coupling, where the curvature $F = \mathrm{d}A + A \wedge A$ fluctuates little around zero, the sum in (36) is dominated by flat connections and their vicinity. Representing flat connections by locally constant transition functions of the principal bundle encodes the gauge field as combinatorial data: we fix a triangulation $\mathcal{M}_{\mathrm{tri}} \simeq \mathcal{M}$, regarded as the nerve of an atlas of local

trivializations of $\mathcal{P}$, and consider simplicial maps $\mathcal{M}_{\rm tri} \to BG_{\rm tri}$. Here $BG_{\rm tri}$ is a simplicial model of the classifying space,

$$\begin{array}{c} EG_{\rm tri} := \bigcup_{n\geq 0} G^{n+1} \Big/ \sim \\ \Big\downarrow \\ BG_{\rm tri} := \bigcup_{n\geq 0} G^{n} \Big/ \sim \end{array} \tag{37}$$

where the simplices in $EG_{\rm tri}$ have their vertices labeled by G, while the simplices in $BG_{\rm tri}$ have their edges labeled by G.

Both are nerves. $BG_{\rm tri} = N_\bullet \mathcal{B}G$ where $\mathcal{B}G$ has one object and $\mathrm{Hom}(*,*) = G$, so an n-simplex is a composable string of n arrows and the labels sit on edges. $EG_{\rm tri} = N_\bullet \,\mathrm{Pair}(G)$ where $\mathrm{Pair}(G)$ has object set G and exactly one arrow between any two objects, so an n-simplex is an $(n{+}1)$-tuple of objects and the labels sit on vertices. $|EG_{\rm tri}|$ is contractible because a groupoid in which every pair of objects is uniquely isomorphic is equivalent to the trivial one. Left multiplication makes G act freely on $\mathrm{Pair}(G)$ with quotient $\mathcal{B}G$, and the functor $(g_0, g_1) \mapsto g_0^{-1} g_1$ is the combinatorial realization of the universal bundle $EG \to BG$ used at the start of §2.3.

A simplicial map $f_{\rm tri} : \mathcal{M}_{\rm tri} \to BG_{\rm tri}$ consists of the data $\{g_{ij} \in G\}$ of 'transition functions' assigned to the edges of $\mathcal{M}_{\rm tri}$, and it determines a principal G-bundle $\mathcal{P}$ over $\mathcal{M}$ together with a flat connection. Flatness is automatic, because the simplicial model built from $\bigcup_n G^n$ with no topology on G is the classifying space of the *discretized* group,

$$BG_{\rm tri} \;\simeq\; BG^\delta, \qquad [\mathcal{M}, BG^\delta] = \{\text{flat } G\text{-bundles}\}/\!\simeq, \tag{38}$$

where G^δ is G with the discrete topology. For continuous G the comparison map $BG^\delta \to BG$ is not an equivalence, so $BG_{\rm tri}$ must not be identified with BG.

For G finite one has $G^\delta = G$, and (36) needs no weak-coupling caveat: there is no curvature to fluctuate, the pair $(\mathcal{P}, A)$ is a single datum, and the sum over $\mathrm{Map}[\mathcal{M}, B_\nabla G]$ is a finite sum over $[\mathcal{M}, BG]$. This is the case the rest of §2 is written for.

Gauge invariance. The relation defining isomorphism classes of Čech data is the *local* coboundary

$$g_{ij} \;\longmapsto\; h_i \, g_{ij} \, h_j^{-1}, \qquad \{h_i\} \in G^{|V|}, \tag{39}$$

of which global conjugation $g_{ij} \mapsto h g_{ij} h^{-1}$ is the special case $h_i \equiv h$. Invariance under global conjugation alone is strictly weaker, and for abelian G it is vacuous: on $\mathcal{M} = \mathbb{S}^1$ triangulated by three edges with $G = \mathbb{Z}_2$, the functional $F[\{g\}] = \#\{e : g_e = -1\}$ is conjugation invariant, yet $h_2 = -1$ carries $(-1,-1,+1)$ to $(+1,+1,+1)$ and F from 2 to 0 at fixed holonomy.

Global conjugation is what remains after gauge fixing. Fixing a spanning tree and using $\{h_i\}$ to set $g_e = e$ on every tree edge forces $h_i = h_j$ along the tree, so the residual gauge freedom is exactly the constant maps: tree-gauge-fixed cocycles are holonomy representations,

and global conjugation is the "$/G$" of $\mathrm{Loc}_G(\mathcal{M}) = \mathrm{Hom}(\pi_1[\mathcal{M}], G)/G$. The two descriptions are related by

$$\frac{|Z^1(\mathcal{M}; G)|}{|G|^{|V|}} \;=\; \frac{|\,\mathrm{Hom}(\pi_1[\mathcal{M}], G)|}{|G|}, \tag{40}$$

numerator and denominator each shrinking by the $|G|^{|V|-1}$ consumed in the gauge fixing. For $G = S_3$ on a 2×2 square-lattice torus one finds $|Z^1| = 3888$ and $|G|^{|V|} = 1296$ on the left, against 18 commuting pairs and $|G| = 6$ on the right, both sides equal to 3.

Gauge invariance of the action need not be imposed as a hypothesis. If the action is built from a cocycle $\omega \in Z^d(BG; \mathbb{R}/\mathbb{Z})$ then (39) changes $f^*\omega$ by a coboundary, and on a closed $\mathcal{M}$ its evaluation vanishes, $\langle \delta\mu, [\mathcal{M}] \rangle = \langle \mu, \partial[\mathcal{M}] \rangle = 0$. Invariance under global conjugation is the special case, inner automorphisms acting trivially on $H^*(BG; \mathbb{R}/\mathbb{Z})$. Accordingly the partition function is a sum over cocycles divided by the gauge volume,

$$\mathcal{Z} = \frac{1}{|G|^{|V|}} \sum_{\{g_{ij}\} \in Z^1(\mathcal{M}; G)} e^{iS[\{g_{ij}\}]} \;=\; \sum_{[\mathcal{P}]} \frac{1}{|\,\mathrm{Aut}\,\mathcal{P}|}\, e^{iS[\mathcal{P}]}, \tag{41}$$

the second form being (44) of §2.4: the summand is constant on gauge orbits, so the sum collapses onto isomorphism classes, each weighted by the reciprocal of its stabilizer. The stabilizer is the centralizer of the holonomy image and varies from orbit to orbit, which is why the weight is needed: in the S_3 example the left side of (40) is 3, the groupoid cardinality, and not the number 8 of isomorphism classes. When $\partial\mathcal{M} \neq \emptyset$ the vanishing above fails, and that failure is the boundary anomaly of §2.3(b).

2.2 Vertex versus edge labelings

The two nerves of §2.1 correspond to the two standard resolutions computing $H^*(BG; \mathbb{R}/\mathbb{Z})$:

$$\omega(g_1, \ldots, g_n) = \widetilde{\omega}(e,\, g_1,\, g_1g_2, \ldots, g_1 \cdots g_n), \qquad \widetilde{\omega}(g_0, \ldots, g_n) = \omega(g_0^{-1}g_1,\, g_1^{-1}g_2, \ldots), \tag{42}$$

with $\widetilde{\omega}$ homogeneous (vertex labels, EG) and ω inhomogeneous (edge labels, BG). The dictionary is

	vertex $\{g_i\}$	edge $\{g_{ij}\}$
Čech degree	0-cochain	1-cochain
symmetry	$g_i \mapsto hg_i$, one h: *on-site global*	$g_{ij} \mapsto h_i g_{ij} h_j^{-1}$: *local redundancy*
topology seen	none (only coboundaries)	$H^1(\mathcal{M}; \underline{G})$
role	matter	background or dynamical gauge field

The map $\{g_i\} \mapsto g_{ij} = g_i^{-1}g_j$ lands in coboundaries only, which is the combinatorial shadow of the contractibility of $|EG_{\mathrm{tri}}|$: a vertex labeling can never produce a nontrivial bundle. The edge data is constrained only by (A) and therefore sees all of H^1.

An SPT phase must use the vertex labeling. Its defining requirement is an *on-site* global symmetry, without which the equivalence relation by finite-depth circuits of individually symmetric gates is not available, and only the vertex labeling carries one. Edge variables carry a redundancy, not a symmetry (Fig. 10).

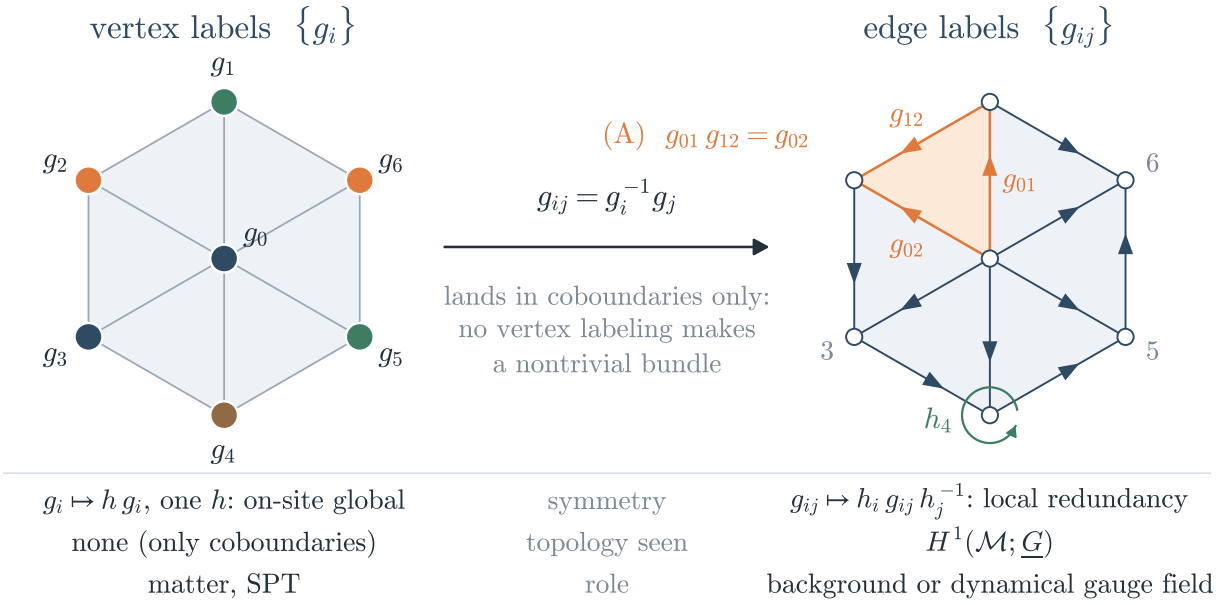


Figure 10: The two labelings of one triangulated patch, with the dictionary of the table below each. Vertex labels (left) carry an on-site global symmetry and, through $g_{ij} = g_i^{-1} g_j$, produce only coboundaries. Edge labels (right) carry a local redundancy h_i at each vertex and are constrained only by the cocycle condition (A) on each 2-simplex (orange; the arrows are the branching structure).

Two technical points make the vertex construction well posed. First, evaluating a cochain on a simplex requires a total order on its vertices (a branching structure); different choices change the action by a coboundary and hence not the cohomology class. Second, the local Hilbert space at each vertex must have a basis labeled by G itself, which is to say the group algebra $\mathbb{C}[G] = \mathrm{span}\{|g\rangle\}$ with G acting by left multiplication (Remark 2.1). Relatedly, the wavefunction of a d-dimensional SPT phase on a closed $(d-1)$-dimensional space Σ is written with an auxiliary reference label g_*: the vertex labels on Σ together with g_* at an apex form a d-chain $C\Sigma$ with $\partial(C\Sigma) = \Sigma$ on which a d-cochain can be evaluated (Fig. 11). Because vertex labels define a map to the contractible EG, every filling gives the same value and the result is independent of g_*. With a nontrivial background on Σ the cone is unavailable: a filling must carry the bundle, need not exist, and two fillings differ by the evaluation of ω on a closed d-manifold. This is the combinatorial counterpart of the bounding-manifold ambiguity of Remark 1.6.

Remark 2.1 (Why the group algebra)**.** *The choice $\mathbb{C}[G]$ is a convenience, not a necessity. It is the local Hilbert space whose basis is a G-torsor, $h|g\rangle = |hg\rangle$, and that is what makes the dictionary exact: vertex labels are then simplicial maps into EG, the symmetry acts freely on them, and the symmetry condition on the wavefunction reads*

$$\Psi[\{hg_i\}] = \Psi[\{g_i\}] \quad \Longleftrightarrow \quad \widetilde{\omega}(hg_0, \ldots, hg_d) = \widetilde{\omega}(g_0, \ldots, g_d),$$

i.e. "symmetric wavefunction" and "homogeneous cochain", hence cochain on $EG/G = BG$, are the same condition. Other on-site representations give the same classification once ancillas

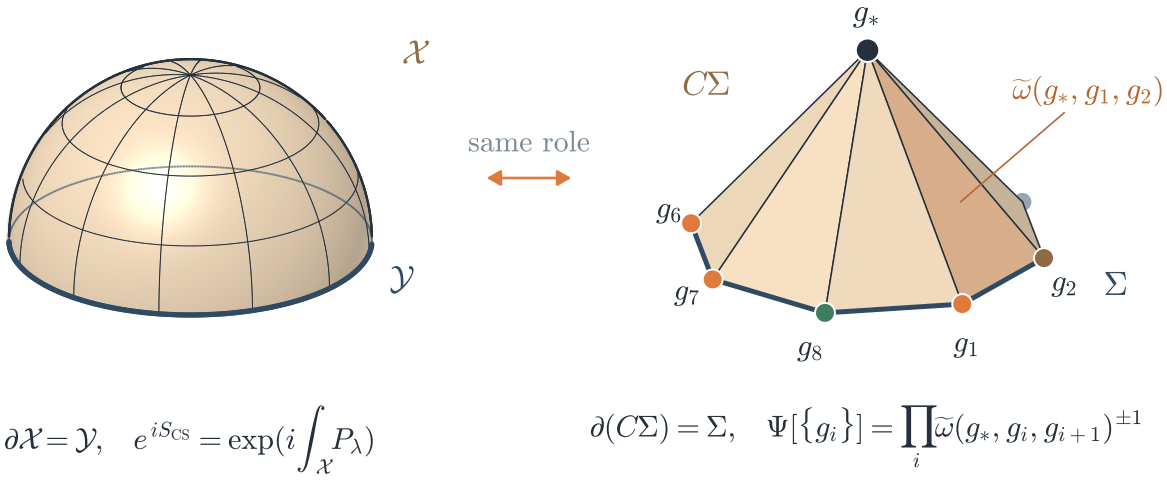


Figure 11: The reference label g_* as a cone. Left: the Chern–Simons action on $\mathcal{Y}$ is computed on a filling $\mathcal{X}$ with $\partial\mathcal{X} = \mathcal{Y}$. Right: the wavefunction of a d-dimensional SPT phase on Σ^{d-1} (here $d = 2$, Σ a circle of eight sites) is computed on the cone $C\Sigma$ with apex g_*, one factor $\widetilde{\omega}(g_*, g_i, g_{i+1})^{\pm 1}$ per d-simplex; on closed Σ the cocycle condition makes the product independent of g_*, and homogeneity of $\widetilde{\omega}$ makes it symmetric.

are allowed; in $\mathbb{C}[G]$ *every class has the explicit fixed-point representative* $\Psi = \prod \widetilde{\omega}^{\pm 1}$. *The decomposition* $\mathbb{C}[G] \simeq \bigoplus_i V_i \boxtimes V_i^*$ *plays no role here; after gauging,* $\mathbb{C}[G]$ *reappears as the link Hilbert space, where this decomposition is the electric-flux basis.*

For $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ *and* $d = 2$ *the cluster chain is an instance: grouping two sites into a cell gives* $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_2] = \mathbb{C}[G]$ *with* η^e, η^o *acting by left multiplication, and the amplitudes of the cluster state factorize as an on-site factor times* $\prod_i \omega(g_i, g_{i+1})$ *with* $\omega(g, g') = (-1)^{ba'}$ *for* $g = (a, b)$, *whose commutator pairing* $\omega(g, g')/\omega(g', g)$ *is* -1 *on the two generators. So* $[\omega]$ *is the nontrivial class in* $H^2(B\mathbb{Z}_2 \times B\mathbb{Z}_2; \mathbb{R}/\mathbb{Z}) = \mathbb{Z}_2$.

2.3 From SPT to gauge theory

Section 1 identified the anomaly of a theory with the response of an invertible bulk. Both are statements about gauging: an anomaly is the obstruction to promoting the background $\mathcal{P}$ to a summed variable, and inflow says that the obstruction disappears for the bulk–boundary system. This section makes the statement concrete for finite G, with dimensions shifted by one: the d-dimensional SPT phase of class $\omega \in H^d(BG; \mathbb{R}/\mathbb{Z})$ plays the bulk, its boundary the anomalous theory. One cocycle then gives three facts: the closed bulk can be gauged, the boundary alone cannot, and gauging the bulk yields $DW_\omega(G)$, whose vacua or anyons record ω although the ungauged partition function is trivial. Gauging is thus both the test of an anomaly and the means of detecting an SPT phase.

We now relate the gauge theory above to another nonlinear sigma model, this time with target G. Recall that a map $\mathcal{M} \to BG$ is equivalent to an equivariant map $\mathcal{P} \to EG$, where

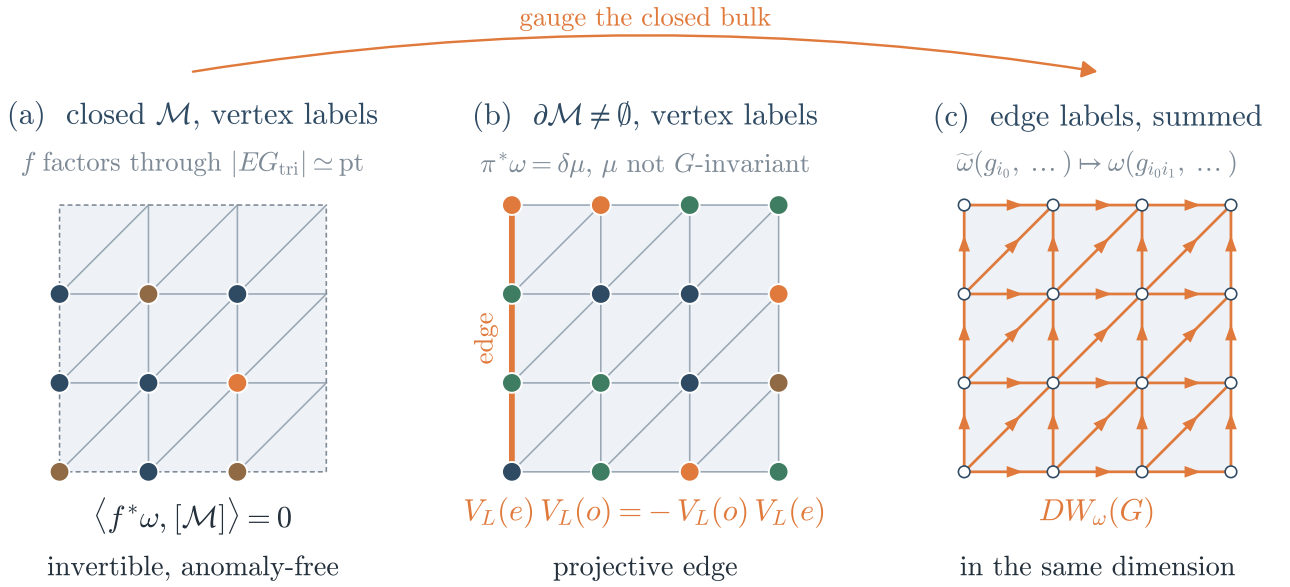


Figure 12: One cocycle ω, three roles. (a) On a closed $\mathcal{M}$ (dashed sides identified) with vertex labels the classifying map factors through the contractible $|EG_{\mathrm{tri}}|$ and the action vanishes. (b) With a boundary (orange) the primitive μ fails to be G-invariant and G acts projectively on the edge; for the cluster chain the edge operators anticommute and the boundary alone cannot be gauged. (c) With edge labels summed over, (43) turns ω into the Dijkgraaf–Witten action. Gauging takes the closed bulk (a) to (c).

$\mathcal{P}$ is a principal G-bundle over $\mathcal{M}$. We replace it by a simplicial map $\mathcal{P}_{\mathrm{tri}} \to EG_{\mathrm{tri}}$ with a triangulation of $\mathcal{P}$ compatible with that of $\mathcal{M}$. One cocycle $\omega \in Z^d(BG; \mathbb{R}/\mathbb{Z})$ then plays three roles, distinguished by which labeling it is evaluated on and whether $\mathcal{M}$ has a boundary (Fig. 12).

(a) Closed $\mathcal{M}$, vertex labeling. The classifying map factors through the contractible $|EG_{\mathrm{tri}}|$, so $[f^*\omega] = 0$ and $\langle f^*\omega, [\mathcal{M}]\rangle = 0$. With no background gauge field the partition function of an SPT phase on a closed spacetime is therefore trivial: the phase is trivial once the symmetry is ignored. The symmetry is on-site by §2.2, so the bulk is anomaly-free and nothing obstructs gauging it.

(b) $\mathcal{M}$ with boundary, vertex labeling. Pulled back to EG one has $\pi^*\omega = \delta\mu$, but the $(d-1)$-cochain μ is not G-invariant, and its failure of invariance is the anomaly of the boundary theory, with ω its inflow. For $d = 2$ this appears as a projective action of G on the edge: for the cluster chain the endpoint operators satisfy $V_L(e)V_L(o) = -V_L(o)V_L(e)$, generating the Pauli algebra $M_2(\mathbb{C}) = \mathbb{C}^\omega[G]$. Gauging the boundary *alone* is then impossible: invariance would require $V_L(e)|\psi\rangle = V_L(o)|\psi\rangle = |\psi\rangle$, which anticommuting operators do not allow, so the projector onto invariant states is identically zero. Equivalently, no trivial summand occurs in an ω-projective representation with $[\omega] \neq 0$.

(c) Edge labeling. Held fixed, the edge labels are a background field and ω is the response of the SPT phase; summed over, they turn ω into the action of a Dijkgraaf–Witten theory. Gauging carries (a) into (c): the vertex variables are gauge-fixed to $g_i = e$, the edge variables become independent subject only to (A), and homogeneity rewrites the weight through (42) as

$$\widetilde{\omega}(g_{i_0}, \dots, g_{i_d}) \;\longmapsto\; \omega(g_{i_0 i_1},\, g_{i_1 i_2}, \dots), \tag{43}$$

whose right-hand side is the Dijkgraaf–Witten action. So gauging a d-dimensional G-SPT phase of class ω yields the twisted theory $DW_\omega(G)$ in the same dimension [3]. It is the closed bulk of (a) that is gauged, not the boundary of (b); a system with boundary can be gauged only as a whole, the inflow of (b) canceling the boundary anomaly.

What $DW_\omega(G)$ is depends on the dimension. For $d = 3$ it is the twisted quantum double $D^\omega(G)$, with genuine topological order; gauging the Levin–Gu phase gives the double semion model. For $d = 2$ there is no topological order and a two-dimensional TQFT is only a collection of vacua, here the irreducible representations of the twisted group algebra $\mathbb{C}^\omega[G]$. For $G = \mathbb{Z}_2 \times \mathbb{Z}_2$: trivial ω gives $\mathbb{C}^4$ with four vacua, and the gauged trivial paramagnet indeed breaks $\hat{G}$ with a fourfold degenerate ground state; nontrivial ω gives $\mathbb{C}^\omega[G] = M_2(\mathbb{C})$ with a single vacuum, matching the unique ground state of the gauged cluster model. That single vacuum is the same two-dimensional projective representation that appeared at the boundary in (b).

Thus *gauging converts a phase distinction with no local order parameter into ordinary data of the gauged theory*: the vacuum structure for $d = 2$, the anyon content for $d = 3$. It does not refine the labeling: by (35) the label ω is the same before and after, and where a labeling by phases does lose information, in the continuous directions of continuous G, the Dijkgraaf–Witten construction is not available, since it presupposes $|G| < \infty$.

2.4 Twisted gauging as stacking with an SPT

§2.3 gauged an SPT phase by itself. More generally, let T be a d-dimensional theory whose G-symmetry is anomaly-free, so that it can be coupled gauge-invariantly to a background $\mathcal{P}$. The coupling is not unique, and the freedom is to stack T with a d-dimensional SPT phase before summing over $\mathcal{P}$: in §1 an invertible phase one dimension up measured the obstruction to the coupling, and here an invertible phase in the same dimension measures the choice of it.

For G finite, $\mathrm{Bun}_G(\mathcal{M})$ is a groupoid, with bundles as objects and gauge transformations as morphisms, so a sum over its objects is weighted by automorphisms as in (41):

$$\mathcal{Z}_{\text{gauged}}[\mathcal{M}] \;=\; \sum_{[\mathcal{P}] \in \pi_0 \mathrm{Bun}_G(\mathcal{M})} \frac{1}{|\operatorname{Aut} \mathcal{P}|}\, \mathcal{Z}[\mathcal{M}; \mathcal{P}]. \tag{44}$$

For trivial $\mathcal{Z}$ this is the untwisted Dijkgraaf–Witten partition function [3]. For $\mathcal{M}$ connected and G abelian one has $\operatorname{Aut} \mathcal{P} \simeq H^0(\mathcal{M}; G) = G$ for every $\mathcal{P}$, so the prefactor is the familiar $1/|G|$: the globally constant gauge transformation cannot be used to gauge-fix, and in the operator formalism it appears as the projector onto the G-invariant sector. For non-abelian G the weight varies from class to class, as in the S_3 example of §2.1.

One may now deform (44) by a bundle-dependent phase $\chi_\omega[\mathcal{P}]$,

$$\mathcal{Z}^{\omega}_{\text{gauged}}[\mathcal{M}] \;=\; \sum_{[\mathcal{P}]} \frac{\chi_\omega[\mathcal{P}]}{|\operatorname{Aut}\mathcal{P}|}\,\mathcal{Z}[\mathcal{M};\mathcal{P}], \qquad |\chi_\omega[\mathcal{P}]| = 1, \tag{45}$$

and this is what is meant by *twisted gauging*: the configurations summed over are unchanged; only the weight attached to each sector changes. The phase is constrained. It must depend only on the isomorphism class of $\mathcal{P}$, or the sum is not gauge invariant; and it must factorize under a decomposition $\mathcal{M} = \mathcal{M}_1 \cup_\Sigma \mathcal{M}_2$, or the gauged theory fails to be local. Together these say that χ_ω is itself the partition function of a d-dimensional invertible theory coupled to the background bundle: in the vocabulary of Definition 1.2, a *topological* action functional and not merely a metric-independent one. The SPT layer of the table in §1 is this topological layer: a Chern–Simons weight depends on a connection and not only on $[\mathcal{P}]$, so it does not belong to it, and for finite G there is none.

Within the ordinary-cohomology ansatz the admissible weights are therefore classified one degree below (5),

$$\chi_\omega[\mathcal{P}] = \exp\left(2\pi i \int_{\mathcal{M}} f_{\mathcal{P}}^{*}\omega\right), \qquad \omega \in H^d(BG;\mathbb{R}/\mathbb{Z}), \tag{46}$$

with $f_{\mathcal{P}} : \mathcal{M} \to BG$ the classifying map of §2.1; the general classification of invertible theories requires the appropriate tangential and bordism data [4, 10]. The decomposition (6), applied in degree d rather than $d+1$, reads

$$H^d(BG;\mathbb{R}/\mathbb{Z}) \simeq \underbrace{(\mathbb{R}/\mathbb{Z})^{b_d}}_{\theta\text{-angles}} \oplus \underbrace{\operatorname{Tor}[H^{d+1}(BG;\mathbb{Z})]}_{\text{discrete torsion}}, \tag{47}$$

so the continuous/discrete dichotomy of §1.1.2 repeats one degree down; for G finite $b_d = 0$ by Note 1.4, leaving discrete torsion alone.

This is the layer in which SPT phases live. The group $H^d(BG;\mathbb{R}/\mathbb{Z})$ is the group-cohomology classification of d-dimensional bosonic SPT phases with symmetry G; for $d = 2$ and $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ it is $\mathbb{Z}_2$, the trivial and cluster phases. Equation (45) gives the classification an operational meaning: an SPT phase is a weight one may insert into a gauging sum.

Remark 2.2 (Two roles of an SPT phase)**.** *Stacking multiplies partition functions, and substituting into* (44) *gives*

$$\mathcal{Z}_{T\otimes \mathrm{SPT}_\omega}[\mathcal{M};\mathcal{P}] = \mathcal{Z}_T[\mathcal{M};\mathcal{P}]\,\chi_\omega[\mathcal{P}], \qquad T/\!\!/^{\omega} G \;=\; (T\otimes \mathrm{SPT}_\omega)/G;$$

for trivial T this is $DW_\omega(G)$ of §2.3. An SPT phase therefore enters in two degrees. One dimension up, a class in $H^{d+1}(BG;\mathbb{R}/\mathbb{Z})$ is the obstruction to coupling T gauge-invariantly to $\mathcal{P}$ (§1). In the same dimension, once that obstruction vanishes, the gauge-invariant couplings form a torsor over $H^d(BG;\mathbb{R}/\mathbb{Z})$, acting by stacking. Stacking cannot remove a nonzero anomaly: being gauge invariant, χ_ω drops out of the ratio $\mathcal{Z}_T[A^g]/\mathcal{Z}_T[A]$, whose representative is shifted only by local counterterms that are not *gauge invariant. The two degrees meet at an interface: where T and $T\otimes\mathrm{SPT}_\omega$ are glued along a codimension-one wall, the wall carries the anomaly ω, by (b) of §2.3. This is why gauging detects SPT phases: no local order parameter distinguishes T from $T\otimes\mathrm{SPT}_\omega$, yet their gauged theories differ once the dual symmetry is kept track of.*

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